

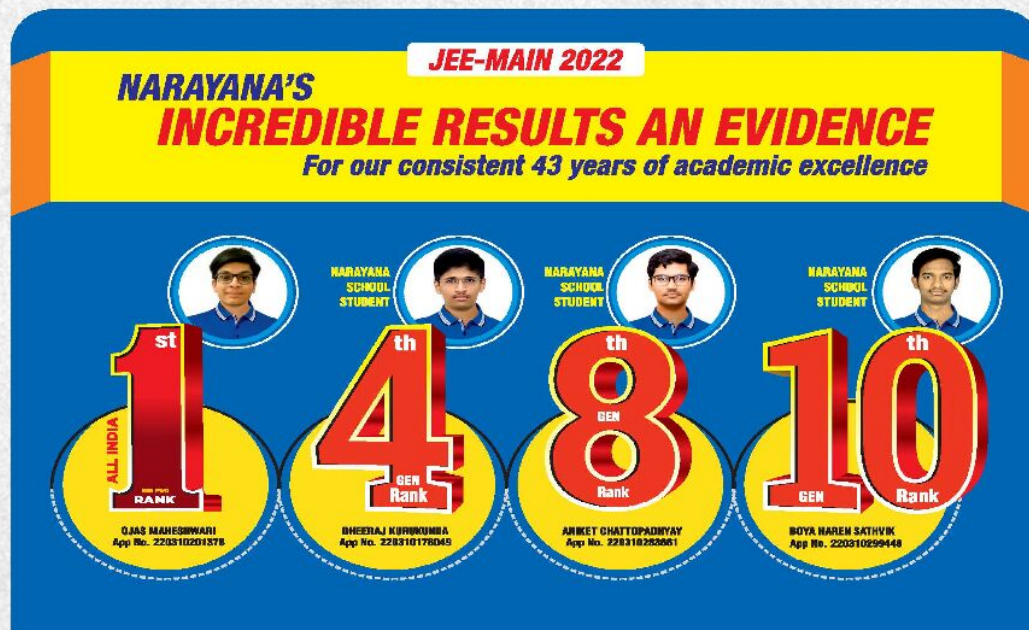


NARAYANA
IIT-JEE ACADEMY - INDIA

NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



ORGANIC CHEMISTRY PART-I (JR) ADVANCED MODEL QUESTIONS



PHYSICS

GRAVITATION, ELECTRICITY
JEE ADVANCED IMPORTANT QUESTIONS

Single correct questions

1. Choose the correct statement(s)
- Work done by magnetic force must always be zero
 - Work done by Ampere force must always be zero
 - Motional emf produces induced electric field
 - Self inductance of an inductor made of super conducting wire must be infinity.

Ans : A

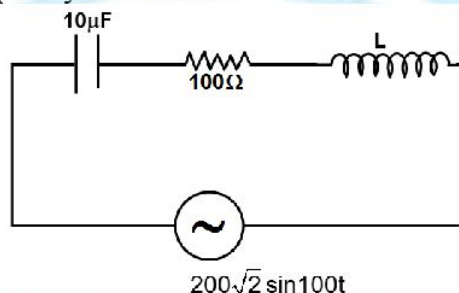
Sol: $\vec{F}_B = q(\vec{v} \times \vec{B})$ always perpendicular to $\vec{v}$

Ampere force is the component of magnetic force perpendicular to the current carrying wire.

Motional emf produces electrostatic field

Inductance is independent of material of the wire

2. An ac source of angular frequency ω is fed across a resistor R and a capacitor C in series. The current registered is I . If now the frequency of source is changed to $\omega/3$ (but maintaining the same voltage), the current in the circuit is found to be halved. The ratio of reactance to resistance at the original frequency ω will be.



(A) $\sqrt{\frac{3}{5}}$

(B) $\sqrt{\frac{5}{3}}$

(C) $\frac{3}{5}$

(D) $\frac{5}{3}$

Ans : A

Sol: According to given problem,

$$I = \frac{V}{Z} = V / [R^2 + (1/C\omega^2)]^{1/2} \dots (1)$$

$$\text{and } \frac{I}{2} = \frac{V}{[R^2 + (3/C\omega)^2]^{1/2}} \dots (2)$$

substituting the value of I from equation (1) in (2),

$$4 \left(R^2 + \frac{1}{C^2\omega^2} \right) = R^2 + \frac{9}{C^2\omega^2}, \quad \text{i.e., } \frac{1}{C^2\omega^2} = \frac{3}{5}R^2$$

$$\text{so that } \frac{X}{R} = \frac{(1/C\omega)}{R} = \frac{[(3/5)R^2]^{1/2}}{R} = \sqrt{\frac{3}{5}}$$

$\therefore$ Correct answer is (A)

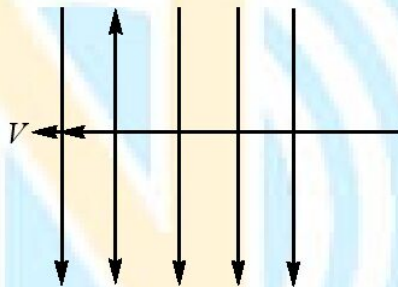
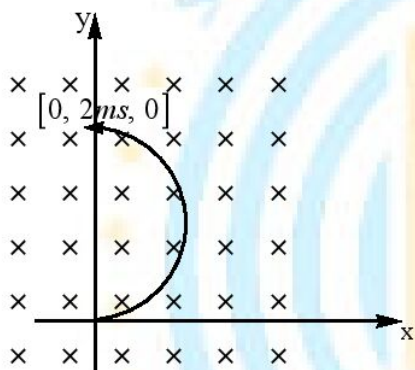
3. The space has electromagnetic field which varies with time whose variation is given as

$$\vec{B} = \begin{cases} -B_0 \hat{k} & \text{if } 0 \leq t \leq \frac{\pi m}{qB_0} \\ -B_0 \hat{j} & \text{if } \frac{\pi m}{qB_0} \leq t \leq \frac{2\pi m}{qB_0} \\ 0 & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if } 0 \leq t \leq \frac{2\pi m}{qB_0} \\ -E_0 \hat{k} & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases}$$

A charge particle having mass 'm' and positive charge 'q' is given velocity $V_0 \hat{i}$ at origin at $t=0$ sec. The coordinate of point on xy plane when it again passes through xy plane for the first time is :

- A) $\left(\frac{2mv_0}{q} \sqrt{\frac{V_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$ B) $\left(\frac{2mv_0}{q} \sqrt{\frac{V_0}{E_0 B_0}}, \frac{mv_0}{qB_0}, 0 \right)$
 C) $\left(\frac{mv_0}{q} \sqrt{\frac{V_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$ D) $\left(\frac{mv_0}{q} \sqrt{\frac{V_0}{E_0 B_0}}, \frac{mv_0}{qB_0}, 0 \right)$

Ans: A
Sol.



F. $t \leq \frac{\pi m}{Bq_0}$

$$\frac{\pi m}{Bq_0} \leq t \leq \frac{2\pi m}{Bq_0}$$

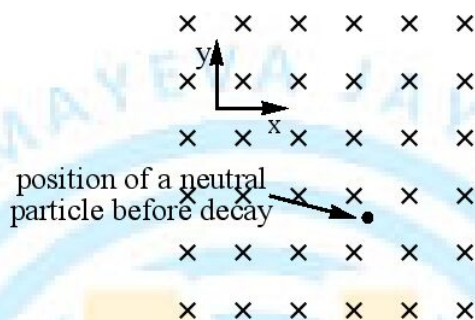
$$t = \frac{2\pi m}{Bq_0} \quad \Delta t = T/2$$

Semi circle centre on axis || Z - axis

$$\left[0, \frac{2mv_0}{Bq}, \frac{2mv_0}{Bq} \right]$$

Hence $\left(\frac{2mv_0}{q} \sqrt{\frac{V_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$

4. A neutral particle is initially at rest in a uniform magnetic field B as shown in the diagram. The particle then spontaneously decays into two fragments, one with a positive charge $+q$ and mass $3m$ and the other with a negative charge $-q$ and mass m . Neglecting the electrostatic & magnetic interactions between the two charged particles and assuming that the speeds are much less than speed of light, the time after the decay at which the two fragments first meet is [Data : $q = 1\mu\text{C}$, $B = 2\pi\mu\text{T}$, $m = 10^{-15}\text{ kg}$ Both the charges have initial velocities in x-y plane]



- A) $250\mu\text{s}$ B) $500\mu\text{s}$ C) $750\mu\text{s}$ D) $1000\mu\text{s}$

Ans: C

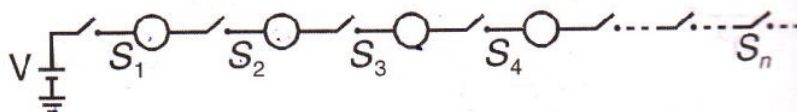
SOL. $r = \frac{P}{qB} = \text{same}, T_+ = \frac{2\pi m_+}{qB} = \frac{6\pi m}{qB}, T_- = \frac{2\pi m}{qB}$

As $T_+ = 3T_-$, they will meet at $\theta = \frac{\pi}{2}$

$q = 1\mu\text{C}, B = 2\pi\mu\text{T}, m = 10^{-15}\text{ kg}$

The time is $\frac{T_+}{4} = \frac{6\pi m}{4qB} = \frac{6 \times \pi \times 10^{-15}}{4 \times 1 \times 10^{-6} \times 2\pi \times 10^{-6}}$
 $= 0.75 \times 10^{-3}\text{ s} = 750\mu\text{s}$

5. All spheres are identical and initially unchanged, S_1 is closed at $t=0$ leaving all others switches open. After sufficient long time S_1 is opened and S_2 is closed. The process is done till S_{n-1} is opened and S_n is closed. Net loss of energy of system if it is made of infinite number of spheres



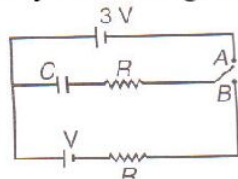
- A) $\frac{5}{6} 4\pi\epsilon_0 R V^2$ B) $\frac{1}{2} 4\pi\epsilon_0 R V^2$ C) $\frac{2}{3} 4\pi\epsilon_0 R V^2$ D) Zero.

Ans:

SOL. $C = 4\pi\epsilon_0 R$

$$AU_{Cons} = \frac{1}{2}CV^2 + \frac{1}{2}C\left(\frac{V}{2}\right)^2 + \frac{1}{2}C\left(\frac{V}{4}\right)^2 + \dots$$

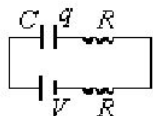
6. A capacitor is initially connected to a battery of EMf 3 V. At $t=0$, switch is thrown to B state. Now charge on capacitor at any instant is given by



- A) $q = C(V + 2Ve^{-t/RC})$ B) $q = C(V - 2Ve^{-t/RC})$
 C) $q = C(V + 2Ve^{-t/2RC})$ D) $q = C(V + 2Ve^{-2t/RC})$

Ans: C

SOL:



$$\frac{q}{C} + 2PR = V$$

$$\Rightarrow Q = \frac{dq}{dt} = \left(V - \frac{q}{C}\right)$$

Solving for $q: 3CV \rightarrow q$ or $t: 0 \rightarrow t$

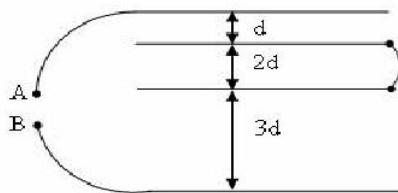
7. When 6 identical cells of no internal resistance are connected in series in the secondary circuit of a potentiometer, the balancing length is '1', balancing length becomes 1/3 when some cells are connected wrongly, the number of cells connected wrongly are

- A) 10 B) 3 C) 2 D) 4

Ans: C

Sol. $E = \rho \ell \Rightarrow 6E_0 = \rho \ell$ and $(6 - 2n)E_0 = \rho \frac{\ell}{3} \Rightarrow n = 2$

8. If the area of each plate is S and the successive separations are d, 2d and 3d then the equivalent capacitance across A and B is :



- A) $\frac{\epsilon_0 A}{6d}$ B) $\frac{\epsilon_0 A}{4d}$ C) $\frac{3\epsilon_0 A}{4d}$ D) none of these

Ans: B

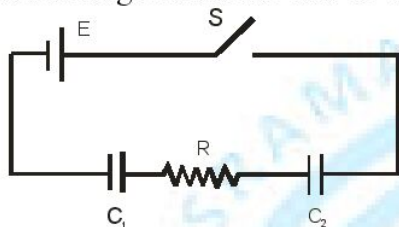
Sol. Between A and B two capacitor (with separation d and 3d) are in series

$$C = \frac{C_1 C_2}{C_1 + C_2}$$

Hence,

$$\frac{\left(\frac{\epsilon_0 A}{d}\right)\left(\frac{\epsilon_0 A}{3d}\right)}{\left(\frac{\epsilon_0 A}{d}\right) + \left(\frac{\epsilon_0 A}{3d}\right)} = \frac{\epsilon_0 A}{4d}$$

9. In the circuit shown in the diagram, $C_1 \neq C_2$. S is a key that is put on at time $t = 0$. Consider the interval during which the charge grows in capacitances C_1 and C_2 . Which of the following statements will be true?



- A) Rate of growth of charge in C_1 and C_2 will be different and the instants at which they are charged maximum will be different.
 B) Rate of growth of charge on C_1 and C_2 will be same only when $C_1 = C_2$
 C) Time constants for C_1 and C_2 will be $\tau_1 = RC_1$ and $\tau_2 = RC_2$ respectively.
 D) Time constants for both C_1 and C_2 will be $\tau = R \left[\frac{C_1 C_2}{C_1 + C_2} \right]$

Ans: D

Sol: The growth of charge in capacitors will be governed as for a single capacitor of value

$$C = \left(\frac{C_1 C_2}{C_1 + C_2} \right) \text{ in the circuit.}$$

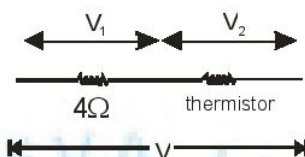
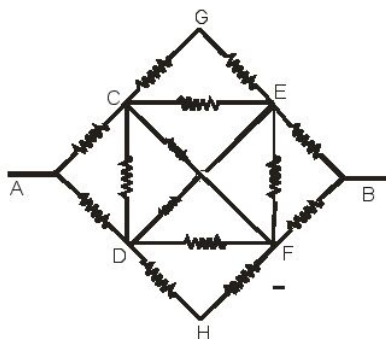
Hence, both capacitors will have the same maximum charge Q , which is reached simultaneously for the two.

10. A 10 V car battery with negligible internal resistance is connected to a series combination of a 4Ω resistor that obeys Ohm's law and a thermistor that does not obey Ohm's law, but instead has a current-voltage relation $V = \alpha I + \beta I^2$ with $\alpha = 2\Omega$ and $\beta = 4\Omega/A$. The current through the 4Ω resistor is

- (A) 1 A (B) 2 A (C) 2/5 A (D) 5 A

Ans: A

Sol: $V = V_1 + V_2$; $10 = 4I + 2I + 4I^2$



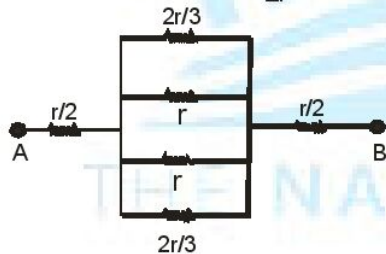
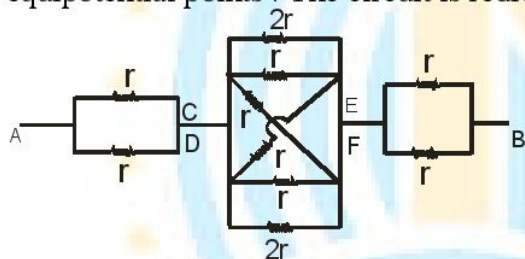
$$4I^2 + 6I - 10 = 0; 2I^2 + 3I - 5 = 0; I = 1A \text{ or } -\frac{5}{2}A$$

11. Fourteen identical resistors each of resistance r are connected as shown. The equivalent resistance between the points A and B is

(A) r (B) $14r$ (C) $\frac{r}{14}$ (D) $1.2r$

ANS:D

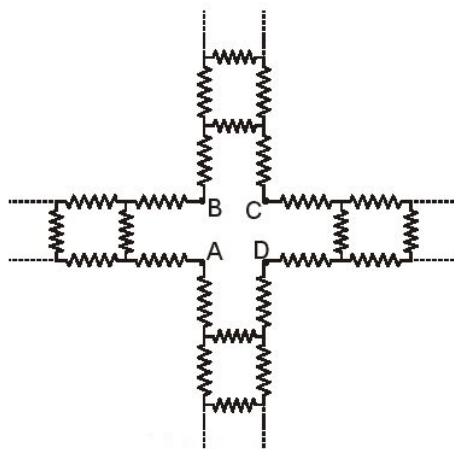
Sol: With respect to the points A and B, the points C and D are equipotential points. E and F are equipotential points. The circuit is redrawn as:



$$\frac{1}{r'} = \frac{2}{r} + \frac{2 \times 3}{2r} = \frac{5}{r}; r' = r/5 \Rightarrow R_{AB} = r' + r = 1.2r$$

12. Four infinite ladder network containing identical resistances of $R \Omega$ each, are combined as shown in figure. The equivalent resistance between A and B is R_{AB} and between A and C is

R_{AC} . Then the value of $\frac{R_{AB}}{R_{AC}}$ is:



(A) $\frac{3}{4}$

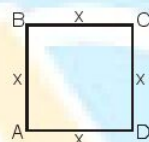
(B) $\frac{4}{2}$

(C) 2

(D) $\frac{1}{2}$

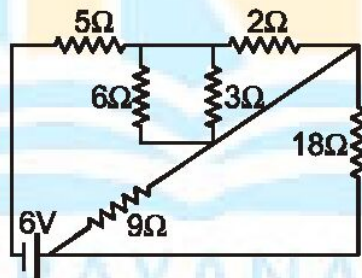
Ans: A

SOL: Let the equivalent resistance of one infinite ladder be x . Then the complete network reduces to



$$\therefore R_{AB} = \frac{x \times 3x}{x + 3x} = \frac{3}{4}x \text{ and } R_{AC} = \frac{2x \times 2x}{2x + 2x} = x \text{ Hence } \frac{R_{AB}}{R_{AC}} = \frac{3}{4}$$

13. In the circuit shown, the cell of emf 6V is ideal. The resistor in which the power dissipated is greatest is



(A) 9 W

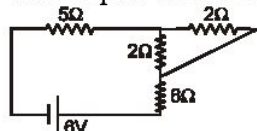
(B) 3 W

(C) 5 W

(D) 18 W

ANS: A

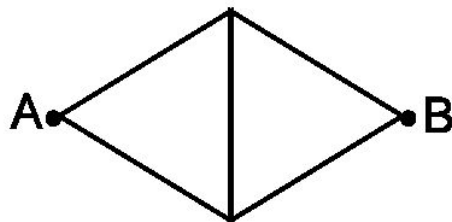
SOL: The equivalent circuit is
The equivalent circuit is



Equivalent resistance across cell = 12 Ω

$\therefore$ Current through cell = $\frac{6}{12} = \frac{1}{2}$ amp. It can be easily verified that power dissipated by 5 Ω resistor is maximum

14. A uniform wire of resistance R is stretched uniformly n times & then cut to form five identical wires. These wires are arranged as shown in the figure. The effective resistance between A & B will be



(A) $\frac{nR}{5}$

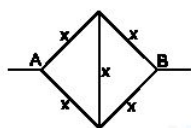
(B) $\frac{R}{5n^2}$

(C) $\frac{n^2R}{5}$

(D) $\frac{n^2R}{2}$

ANS: C

SOL:



Resistance of wire $R = \rho \frac{\ell}{A}$

$\ell A = \text{volume of wire} = \text{constant}$

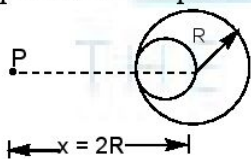
If $\ell \rightarrow n\ell$ then $A \rightarrow \frac{A}{n}$

hence the resistance of wire becomes $R' = n^2 R$

The resistance of each $\frac{1}{5}$ th part is $x = \frac{R'}{5} = \frac{n^2 R}{5}$ This is a balanced wheat stone bridge

$\Rightarrow$ Equivalent resistance across AB is $R_{eq} = x = \frac{n^2 R}{5}$

15. A solid spherical region of radius R having a spherical cavity of diameter R (equal to the radius of the spherical region) has charge of uniform density ρ . The electric field and potential at a point P at $x = 2R$ as shown is



(A) $\frac{1}{27} \left(\frac{\rho R}{\epsilon_0} \right), \frac{5}{12} \left(\frac{\rho R^2}{\epsilon_0} \right)$

(B) $\frac{7}{108} \left(\frac{\rho R}{\epsilon_0} \right), \frac{5}{36} \left(\frac{\rho R^2}{\epsilon_0} \right)$

(C) $3 \left(\frac{\rho R}{\epsilon_0} \right), \frac{5}{6} \left(\frac{\rho R^2}{\epsilon_0} \right)$

(D) $\frac{1}{9} \left(\frac{\rho R}{\epsilon_0} \right), \frac{1}{9} \left(\frac{\rho R^2}{\epsilon_0} \right)$

Ans : B

Sol : Charge density $\rho = \frac{Q}{\frac{4}{3}\pi\left(R^3 - \left(\frac{R}{2}\right)^3\right)} = \frac{6Q}{7\pi R^3}$

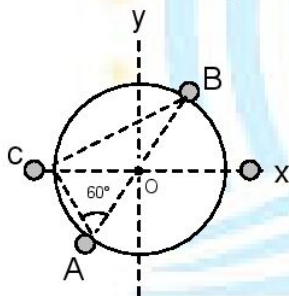
Using the superposition principal $V = \frac{\rho\left(\frac{4}{3}\pi R^3\right)}{4\pi\epsilon_0 X} - \frac{\left(\rho\frac{4}{3}\pi\left(\frac{R}{2}\right)^3\right)}{4\pi\epsilon_0\left(X - \frac{R}{2}\right)} = \frac{\rho R^3}{3\epsilon_0} \frac{(7X - 4R)}{4X(2X - R)}$

At $X=2R$, We get $V = \frac{5}{36} \left(\frac{\rho R^2}{\epsilon_0} \right)$

Similarly, $E = \frac{\rho\left(\frac{4}{3}\pi R^3\right)}{4\pi\epsilon_0 X^2} - \frac{\rho\left(\frac{4}{3}\pi\left(\frac{R}{2}\right)^3\right)}{4\pi\epsilon_0\left(X - \frac{R}{2}\right)^2}$; $E = \frac{\rho R^3}{3\epsilon_0} \left(\frac{1}{X^2} - \frac{1}{2(2X - R)^2} \right)$

So, at $X=2R$, we get $E = \frac{2\rho R}{27\epsilon_0} = \frac{2}{27} \left(\frac{\rho R}{\epsilon_0} \right)$

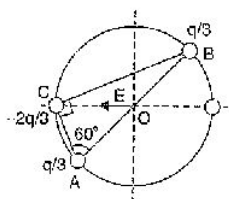
16. Consider a system of three charges $\frac{q}{3}$, $\frac{q}{3}$ and $-\frac{2q}{3}$ placed at points A, B and C, respectively, as shown in the figure. Take O to be the centre of the circle of radius R and Angle $CAB = 60^\circ$.



- (A) The electric field at point O is $\frac{q}{8\pi\epsilon_0 R^2}$ directed along the negative x -axis
- (B) The potential energy of the system is zero
- (C) The magnitude of the force between the charges of C and B is $\frac{q^2}{54\pi\epsilon_0 R^2}$
- (D) The potential at point O is $\frac{q}{12\pi\epsilon_0 R}$

Ans : C

SOL: Net electric field due to both charged $\frac{q}{3}$, will get cancelled. Electric field due to $\left(-\frac{2q}{3}\right)$ will be directed in $-ve$ axis



$$E = \frac{1}{4\pi\epsilon_0} \frac{\left(\frac{2q}{3}\right)}{R^2} \Rightarrow E = \frac{q}{6\pi\epsilon_0 R^2}$$

$$\text{P.E of system} = \frac{1}{4\pi\epsilon_0} \left(\frac{\left(\frac{q}{3}\right)^2}{2R} + \frac{\frac{q}{3} \left(-\frac{2q}{3}\right)}{2R \sin 60^\circ} + \frac{\frac{q}{3} \left(-\frac{2q}{3}\right)}{2R \cos 60^\circ} \right)$$

P.E of system $\neq 0$

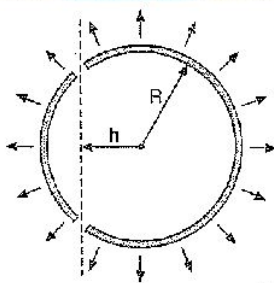
Force between B and C

$$F = \frac{1}{4\pi\epsilon_0} \frac{\left(\frac{2q}{3}\right) \left(\frac{q}{3}\right)}{(2R \sin 60^\circ)} = \frac{1}{4\pi\epsilon_0} \frac{4 \times 2q^2}{9 \times 4 \times 3R^2} \Rightarrow F = \frac{2q^2}{9 \times 3 \times 4\pi\epsilon_0 R^2} \text{ (attractive)} \Rightarrow F = \frac{1}{54} \frac{q^2}{\pi\epsilon_0 R^2}$$

Potential at O is

$$V = \frac{1}{4\pi\epsilon_0} \frac{\frac{q}{3} + \frac{q}{3} - \frac{2q}{3}}{R} = 0$$

17. Consider a metal sphere, of radius R that is cut in two along a plane whose minimum distance from the sphere's centre is h . Sphere is uniformly charged by a total electric charge Q . The force necessary to hold the two parts of the sphere together is



(A) $F = \frac{Q^2}{4\pi\epsilon_0 R^4} (R^2 - h^2)$

(B) $F = \frac{Q^2}{4\pi\epsilon_0 R^2}$

(C) $F = \frac{Q^2}{32\pi\epsilon_0 R^3} (R - h)$

(D) $F = \frac{Q^2}{32\pi\epsilon_0 R^4} (R^2 - h^2)$

ANS : D

SOL: This electric field exerts a force $\Delta F = \frac{1}{2} E \Delta Q$ on the charge $\Delta Q = \sigma \Delta A$ which resides on a

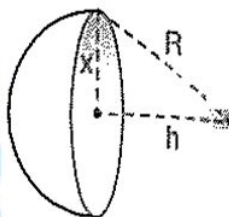
surface of area ΔA , as illustrated in figure. The reason for the factor of $\frac{1}{2}$ is that the electric

field strength is E surface of the surface of the sphere and zero inside and hence its average value comes out to be $\frac{E}{2}$.

The force per unit area exerted by the charges on the pieces of the sphere is therefore

$$\frac{\Delta F}{\Delta A} = \frac{Q^2}{32\pi^2 \epsilon_0 R^4} = P$$

The required force can be considered to be analogous to the force with which a liquid having pressure p would push apart the two pieces of the sphere and this force is also the product of P and the cross-sectional area of the intersection of the plane and sphere.

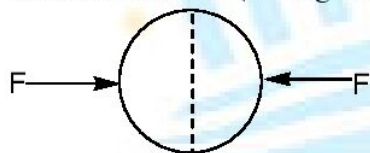


$$\text{So, } F = pa = p(\pi X^2)$$

$\Rightarrow F = p\pi(R^2 - h^2)$ Hence it follows that the two parts of the sphere can be held together by a force

$$F = \frac{Q^2}{32\pi\epsilon_0 R^4} (R^2 - h^2)$$

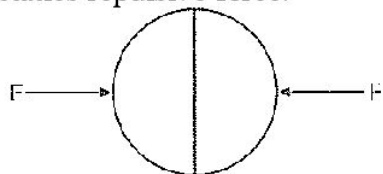
18. A uniformly charged thin spherical shell of radius R carries uniform surface charge density of σ per unit area. If the shell is made of two hemispherical shells, held together by pressing them with force F (see figure), then the minimum value of F should be



- (A) $\frac{\pi}{2\epsilon_0} \sigma^2 R^2$ (B) $\frac{1}{\epsilon_0} \sigma^2 R$ (C) $\frac{1}{\epsilon_0} \frac{\sigma^2}{R}$ (D) $\frac{1}{\epsilon_0} \frac{\sigma^2}{R^2}$

Ans : A

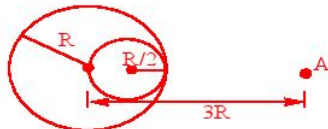
SOL: Electrostatics repulsive force:



$$F_{ele} = \left(\frac{\sigma^2}{2\epsilon_0} \right) \pi R^2 ; \quad F = F_{ele} = \frac{\sigma^2 \pi R^2}{2\epsilon_0}$$

19. A solid sphere of uniform density and radius R applies a gravitational force of attraction

equal to F_1 on a particle placed at a distance $3R$ from the centre of the sphere. A spherical cavity of radius $\frac{R}{2}$ is now made in the sphere as shown in the figure. The sphere with cavity now applies a gravitational force F_2 on the same particle. The ratio $\frac{F_2}{F_1}$ is



A) $\frac{9}{50}$

B) $\frac{41}{50}$

C) $\frac{3}{25}$

D) $\frac{22}{25}$

Ans : B

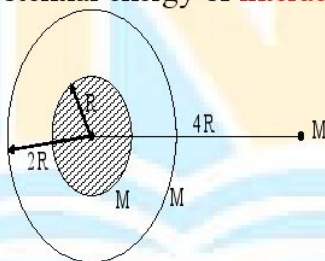
Sol. M = mass of complete sphere $= \frac{4}{3}\pi R^3 \rho$

Mass of the removed part $= \frac{4}{3}\pi R^3 \rho / 8$

$$F_2 = \frac{GMm}{9R^2} - \frac{GMm}{50R^2}$$

$$F_1 = \frac{GMm}{9R^2}$$

20. A solid sphere of mass M and radius R lies concentrically inside a spherical shell of mass M and radius $2R$. A particle of mass M is placed at a distance $4R$ from the common centre as shown in the figure. The total potential energy of interaction of the system of masses is



(A) $-\frac{GM^2}{R}$

(B) $-\frac{GM^2}{2R}$

(C) $-\frac{GM^2}{4R}$

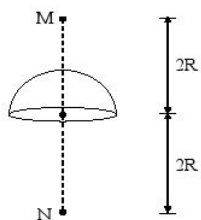
(D) $-\frac{GM^2}{8R}$

Ans : A

Sol. Total potential energy of the system = interaction energy of them.

$$\left[-\frac{GMM}{4R} - \frac{GMM}{4R} - \frac{GMM}{2R} \right] = -\frac{GM^2}{R}$$

21. Two points M and N are situated at a distance $2R$ from the centre (C) of a solid hemisphere of mass M and radius R as shown in the figure. If the magnitude of the gravitational field at N is E . Then the magnitude of gravitational field at M is



- (A) E (B) $\frac{GM}{4R^2} + E$ (C) $\frac{GM}{4R^2} - E$ (D) $\frac{GM}{2R^2} - E$

Ans : D

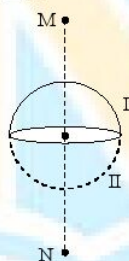
Sol. Let us place another identical hemisphere (II) below the given hemisphere (I) as shown in the figure.

Now the magnitude of gravitational field at N due to II is equal to magnitude of gravitational field at M due to I.

Also the net gravitational field due to hemisphere I and II at M = gravitational field at M due to I + gravitational field at M due to II

$$\text{i.e. } \frac{G(2M)}{(2R)^2} = \text{gravitational field at M due to I} + E$$

$$\therefore \text{gravitational field at M due to I} = \frac{GM}{2R^2} - E$$

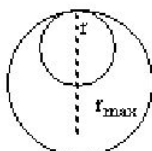


22. A satellite is revolving round the earth in a circular orbit of radius r and velocity V_0 . A particle is projected from the satellite in forward direction with relative velocity $V = \left(\sqrt{\frac{7}{6}} - 1\right)V_0$, then the maximum distance of the particle from the earth's centre during the subsequent motion of the particle is

- A) $\frac{5r}{3}$ b) $\frac{7r}{5}$ C) $\frac{7r}{4}$ D) None

Ans :B

Sol :



$$V_{\text{Absolute}} = V_{R.V} + V_S$$

$$= \left(\sqrt{\frac{7}{6}} - 1 \right) V_0 + V_0 = \sqrt{\frac{7}{6}} V_0$$

$$V_{\text{Absolute}} > V_0$$

Outer ellipse

Apply conservation of angular momenta $Li = Lf$, $\sqrt{\frac{7}{6}} m v_0 r = m v r_{\text{max}}$

$$v_0 = \sqrt{\frac{GM}{r}}$$

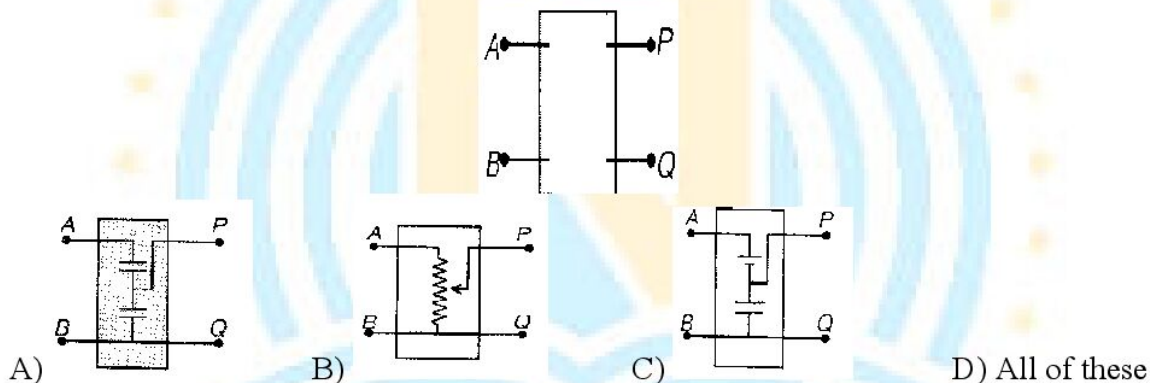
Apply conservation of Total energy

$$\left(\frac{7}{6} \right) \frac{1}{2} m v_0^2 - \frac{GMm}{r} = \frac{1}{2} m v^2 - \frac{GMm}{r_{\text{max}}}$$

Solve the quadratic equation

$$\text{Then } r_{\text{max}} = 7r/5$$

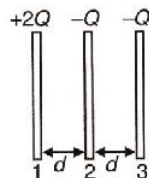
23. If an ideal battery of 10 V is connected across A and B, potential difference measured by voltmeter between P and Q is 10 V. Correct circuit in box is .



ANS: C

Sol: Capacitor charges to the emf of cell

24. Three identical square plates of surface area A of each side is arranged as shown in the figure. Plates 1, 2, and 3 are given charges $+2Q$, $-Q$ and $-Q$ respectively. Find out the correct options

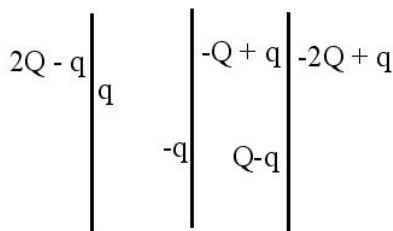


- A) Charge on the two faces of the middle plate are $-\frac{Q}{2}$ and $-\frac{Q}{2}$
 B) Charge on the two faces of the middle plate are $-Q$ and 0
 C) Energy of the system is $\left(\frac{5 Q^2 d}{2 \epsilon_0 A} \right)$

D) Energy of the system is $\left(\frac{3 Q^2 d}{2 \epsilon_0 A} \right)$

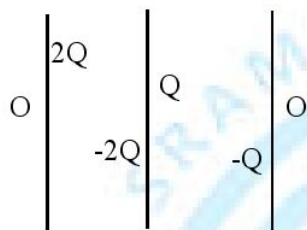
Ans: C

Sol:



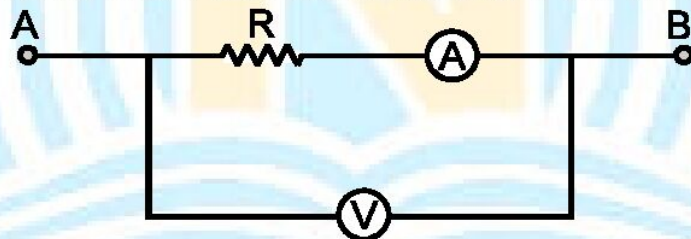
$$2Q - q = -2Q + q$$

$$\Rightarrow q = 2Q$$



$$U = \frac{(2Q)^2}{2C} + \frac{Q^2}{2C} = \frac{5Q^2 d}{2 \epsilon_0 A}$$

25. A non-ideal voltmeter and a non-ideal ammeter are connected as shown in the figure. The reading of the voltmeter is 20 V and that of the ammeter is 4 A. The value of R is:



(A) 5 Ω

(B) less than 5 Ω

(C) greater than 5 Ω

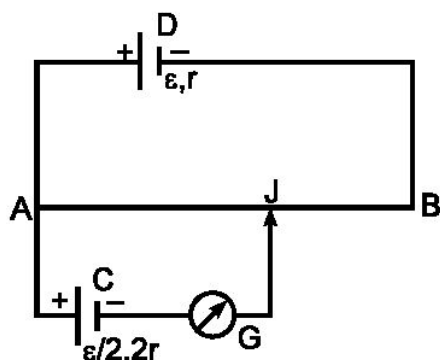
(D) may be less than 5 Ω

Ans: B

Sol: $(R + R_a)i = V \Rightarrow R + R_a = \frac{V}{i}$

(or) $R \propto \frac{V}{i}$

26. In the fig. the potentiometer wire AB of length L & resistance 9 r is joined to the cell D of e.m.f. e and internal resistance r. The cell C's e.m.f. is e/2 and its internal resistance is 2 r. The galvanometer G will show no deflection when the length AJ is:



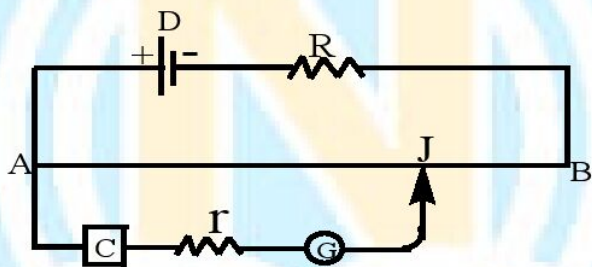
- (A) $4L/9$ (B) $11L/18$ (C) $7L/18$ (D) $5L/9$

Ans: D

Sol: $\rho = \frac{E(qr)}{(qr+r)L} = \frac{9E}{10L}$

$$\frac{E}{2} = \rho \ell \Rightarrow \ell = \frac{5L}{9}$$

27. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0 . 'C' is a cell whose internal resistance r is separately shown. The zero centered galvanometer 'G' does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this?

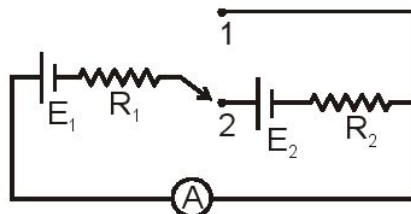


- (A) $r > R_0$ (B) $R \gg R_0$
(C) emf of cell C > emf of cell D
(D) The negative terminal of C is connected to A.

Ans : A

Sol: If $V_{AB} < E_C$ then no zero deflection

28. The resistance R_1 is adjusted in such a way that whether the switch is in position 1 or in 2, the current through the ideal ammeter is $I = 2$ A. The value of emf of $E_2 = 6$ V. The cells are ideal. The value of resistance R_2 is



- (A) Equal to 3Ω (B) greater than 3Ω

(C) Less than 3Ω (D) Dependent on E_1

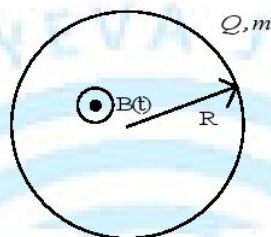
ANS: A

Sol: $E_1 = 2R_1$ and $(E_1 + E_2) = 2(R_1 + R_2)$

$$\Rightarrow R_2 = 3\Omega$$

One or More correct questions

29. A thin non conducting circular ring of mass m , radius R has a uniformly distributed charge Q . The ring is placed on rough horizontal floor of friction coefficient μ . A uniform vertically upward magnetic field of radius greater than R exists coaxial with the ring. The magnetic field is varying as $B = B_0 t^2$ where t is time and B_0 is a positive constant, INCORRECT of the following are



- (a) Induced electric field is $Bt \frac{R}{2}$
 (b) torque applied by the electric force on the ring at any time 't' is : $\frac{B_0 Q R^2 t}{2}$
 (c) the time at which the ring starts rotating is $\frac{\mu mg}{B_0 Q R}$
 (d) the ring would never rotate

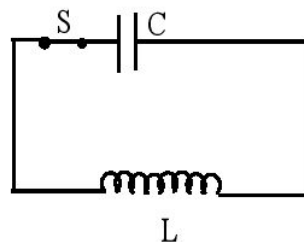
Ans : ABD

Sol: $E = \frac{1}{2} \frac{dB}{dt} \cdot R$

$$\tau = EQR$$

when $\tau = \mu mgR$, the will start rotating

30. In the given circuit, initially the charge on the capacitor is Q_0 . At time $t=0$, the switch (s) is closed. Which of the following statement (s) is/are correct?



- (a) Maximum current through the inductor is $\frac{Q_0}{\sqrt{LC}}$
- (b) At time $t = \frac{\pi}{2}\sqrt{LC}$, charge on the capacitor becomes zero
- (c) At time $t = \frac{\pi}{2}\sqrt{LC}$, current through the inductor becomes maximum
- (d) Maximum magnetic energy stored in the inductor is $\frac{Q_0^2}{2C}$

ABCD

SOL: $I_{\max} = Q_0 \omega$ at $t = \frac{T}{4}$ current becomes maximum and charge on the capacitor becomes zero

Total energy is conserved

31. A particle of specific charge α enters with a velocity $\vec{v} = v_0 \hat{i}$ in a magnetic field

$$\vec{B} = \frac{B_0}{\sqrt{2}}(\hat{j} + \hat{k}). \text{ Then}$$

A) path of the particle is a helix B) path of the particle is a circle

C) distance moved by particle in time $t = \frac{\pi}{B_0 \alpha}$ is $\frac{\pi v_0}{B_0 \alpha}$ D) velocity of particle after time $t = \frac{\pi}{B_0 \alpha}$ is $\left(\frac{v_0}{2} \hat{i} + \frac{v_0}{2} \hat{j}\right)$

Ans: BC

SOL. $\vec{v} \perp \vec{B} \Rightarrow \text{circle}$ Distance moved in $t = R\theta = R\omega t$

$$= vt = \frac{\pi v_0}{B_0 \alpha}$$

$$T = 2\pi \frac{m}{B_0 q} = \frac{2\pi}{B_0 \alpha} \text{ after a time of } \frac{\pi}{B_0 \alpha}$$

The particle comes 180° Thus $\vec{v} = -v_0 \hat{i}$

32. In a region, current density is $\vec{j} = \frac{k}{r^2} \hat{r}$ (k is a positive constant and $\hat{r}$ is a vector in the radially out ward direction from a fixed point O). In this region, due to this current, except at the point O

A) $B = 0$ at every point B) $E \neq 0$ at any point
 C) $B \neq 0$ at any point D) $E = 0$ at every point

Ans: AB

Sol: An electric current needs an electric field. Using ampere's law 'B' can be evaluated

33. Among the following statements the correct statements are

A) Biot – Savart's law can be derived from Ampere's circuital law
 B) Biot – Savart's law cannot be derived from Ampere's circuital law

- C) Ampere's circuital law cannot be derived from Biot – Savart's law
 D) Ampere's circuital law can be derived from Biot – Savart's law

Ans: AD

Sol: Both the laws are respectively magnetic effects of electric current

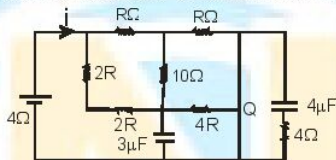
34. A charged particle goes undeflected in a region of space containing an electric field $\vec{E}$ and a magnetic field of induction $\vec{B}$. Both the fields are non zero and uniform. Which of the following is possible?

- a) $\vec{E}$ is parallel to $\vec{B}$ but $\vec{v}$ is perpendicular to $\vec{E}$
 b) $\vec{E}$ is parallel to $\vec{B}$ and $\vec{v}$ parallel to $\vec{E}$
 c) $\vec{E}$ is perpendicular to $\vec{B}$ and $\vec{v}$ is parallel to $\vec{E} \times \vec{B}$
 d) $\vec{E}$ is perpendicular to $\vec{B}$ and $\vec{v}$ is parallel to $\vec{E} \cdot \vec{B}$

Ans: BC

Sol: Net electric force and magnetic force should be parallel to $\vec{v}$

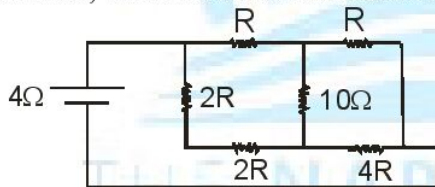
35. A battery of internal resistance 4Ω is connected to the network of resistances as shown. In order that maximum power can be delivered to the network.



- (A) The value of R in ohm should be 2.5Ω
 (B) The value of R in ohm should be 6Ω
 (C) The value of R depends on the voltage of the battery
 (D) The value of R is independent of the voltage of the battery

Sol: In order to deliver maximum power the internal resistance of the source must be equal to external resistance.

Hence, external resistance must be 4Ω

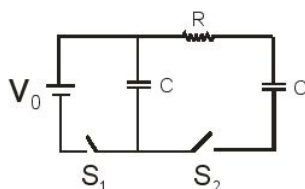


The circuit is a balanced wheatstone's bridge

The effective resistance of the circuit is

$$\left(\frac{2R \times 8R}{2R + 8R} \right) = \frac{8}{5}R ; \therefore \frac{8}{5}R = 4 \Rightarrow R = 2.5\Omega$$

36. In a circuit shown in figure, the capacitance of each capacitor is equal to C and the resistance equal to R. One of the capacitance was connected to a voltage V_0 by closing switch S_1 . Now at $t = 0$, the switch S_1 is opened and S_2 is closed



- A) The current in the circuit as a function of time t is $I = \frac{V_0}{R} e^{-2t/RC}$
- (B) Charge as a function of time is $q = \frac{CV_0}{2} (1 - e^{-2t/RC})$
- (C) Heat produced in the circuit is $\frac{CV_0^2}{2}$
- (D) Heat produced in the circuit is $\frac{CV_0^2}{4}$

Ans: ABD

Sol. Initially charge on capacitor 1 is $Q = CV_0$

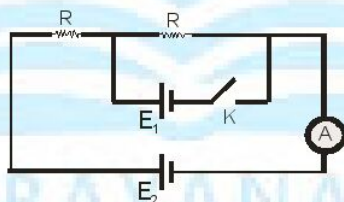
Applying Kirchoff's law :

$$\frac{Q-q}{C} - IR - \frac{q}{C} = 0 \Rightarrow IR = \left(\frac{Q-2q}{C} \right) \Rightarrow \int_0^q \left(\frac{dq}{Q-2q} \right) = \int_0^t \frac{dt}{RC}$$

$$\Rightarrow q = \frac{CV_0}{2} (1 - e^{-2t/RC}) \text{ Hence, } I = \frac{V_0}{R} e^{-2t/RC}$$

$$\begin{aligned} \text{Heat produced } H &= \int_0^\infty i^2 R dt = \int_0^\infty \left(\frac{V_0}{R} e^{-2t/RC} \right)^2 R dt \\ &= \frac{V_0^2}{R} \left[\frac{e^{-4t/RC}}{-4/RC} \right]_0^\infty = \frac{-CV_0^2}{4} [e^{-\infty} - e^{-0}] = \frac{1}{4} CV_0^2 \end{aligned}$$

37. In the given circuit, when key K is open, reading of ammeter is I . Now key K is closed. Then the correct statement is :



- (A) If $E_1 = IR$, reading of the ammeter is I .
- (B) If $IR < E_1 < 2IR$, reading of the ammeter is greater than I .
- (C) If $E_1 = 2IR$, reading of the ammeter will be zero.
- (D) Reading of the ammeter will not change.

Ans: AC

Sol: When K is open current in the ammeter is $I = \frac{E_2}{2R} \Rightarrow E_2 = 2IR$

When switch K is closed.

$$\text{Current in ammeter becomes } I' = \frac{E_2 - E_1}{R}$$

$$\text{When } E_1 = IR \Rightarrow I' = I$$

Hence, (A) Is correct

When $IR < E_1 < 2IR$, reading of the ammeter will be less than I.

$$I' = \left(\frac{2IR - 2IR}{R} \right) = 0$$

Hence, (C) is correct

38. When no current is passed through a conductor, then

(A) The free electrons do not move.

(B) The average speed of a free electron over a large period of time is zero

(C) The average velocity of a free electron over a large period of time is zero

(D) The average of the velocities of all free electrons at an instant is zero

Ans : CD

Sol : When no P.D is applied across a conductor, electron move only by thermal oxidation

39. Three concentric conducting spherical shells A, B and C have radii r , $2r$ and $3r$ and possess charges Q_1 , Q_2 , and Q_3 respectively. The innermost and the outermost shells are earthed individually. Select the mathematical relations between the charges that are correct.

$$(A) Q_1 + Q_3 = -Q_2 \quad (B) Q_1 = -\frac{Q_2}{4} \quad (C) \frac{Q_3}{Q_1} = 3 \quad (D) \frac{Q_3}{Q_2} = -\frac{1}{3}$$

Ans : ABC

SOL. Potential of innermost shell is zero. So,

$$\frac{Q_1}{r} + \frac{Q_2}{2r} + \frac{Q_3}{3r} = 0 \Rightarrow 6Q_1 + 3Q_2 + 2Q_3 = 0$$

Similarly, potential of the outermost shell is also zero. So,

$$\Rightarrow \frac{Q_1}{3r} + \frac{Q_2}{3r} + \frac{Q_3}{3r} = 0 \Rightarrow Q_1 + Q_2 + Q_3 = 0$$

Solving equations (1) and (2) we get

$$Q_1 = \frac{-Q_2}{4}, \frac{Q_3}{Q_1} = 3 \text{ and } \frac{Q_3}{Q_2} = -\frac{3}{4}$$

40. A particle having specific charge s starts from rest in a region where the electric field has constant direction but magnitude varying with time t as $E = E_0 t$. In time t , it is observed that the particle acquires a velocity v after covering a distance x , then (Assume that only force due to the given electric field is acting on the charge)

$$(A) x = \frac{1}{2}(E_0 s)t^3 \quad (B) x = \frac{1}{6}(E_0 s)t^3 \quad (C) v = (E_0 s)t^2 \quad (D) v = \frac{1}{2}(E_0 s)t^2$$

Ans: BD

SOL. $E = E_0 t$

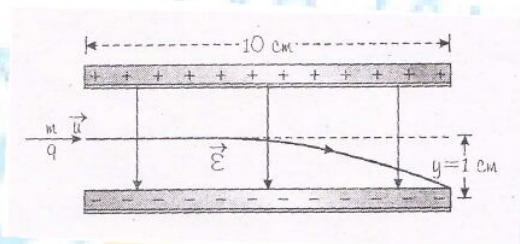
Since

$$a = \frac{qE}{m} = \frac{qE_0 t}{m} = E_0 s t \Rightarrow \frac{dV}{dt} = E_0 s t \Rightarrow dV = E_0 s t dt$$

$$\Rightarrow V = E_0 S \int_0^t t dt = \frac{1}{2} E_0 s t^2 \Rightarrow \frac{dX}{dt} = \frac{1}{2} E_0 s t^2$$

$$\Rightarrow dX = \frac{1}{2} E_0 s t^2 dt \Rightarrow \int_0^X dX = \frac{1}{2} E_0 S \int_0^t t^2 dt \Rightarrow X = \frac{1}{6} E_0 s t^2$$

41. A particle having a charge of 1.6×10^{-19} C enters midway between two parallel plates as shown in the figure. The initial velocity of particle is parallel to the plates. A potential difference of 300 volts is maintained between the plates. If the length of the plates is 10cm and they are separated by 2cm. The mass of the particle is 12×10^{-24} kg. (Consider the electric field to be uniform between the plates)

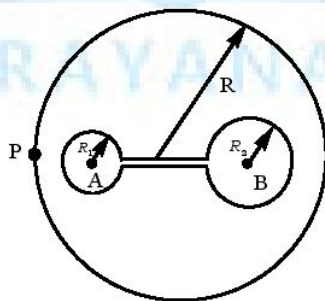


- (A) the particle will be able to come out of the plates for an initial velocity 10^5 m/s
 (B) the particle will be able to come out of the plates for any initial velocity greater than 10^4 m/s
 (C) the particle will be able to come out of the plates for any arbitrary initial velocity
 (D) the particle will not be able to come out of the plates for any initial velocity

Ans: AB

Sol: The particle moves with constant acceleration, Hence applying standard equation, Ans is AB

42. An uncharged spherical conductor of radius $R = 10r$ has two spherical cavities of radii $R_1 = 2r$ and $R_2 = 3r$ connected by an extremely narrow smooth tunnel. A, B are the centers of the cavities as shown. A point charge 'q' is arranged at B. Then it is transferred very slowly from B to A. Which of the following statements is true



- A) Initial and final surface charge densities on the sphere at the point P are same.
 B) Initial and final energy densities at a distance $l (=16r)$ from the centre of the sphere are same
 C) The net electro static potential energy within the cavities increases as the charge is transferred from B to A

D) Work done by the agent in transferring the charge is $\frac{-q^2}{48\pi\epsilon_0 r}$

Ans : ABD

SOL. $r_2 = r + ba$ ($b = \text{constant}$)

$$E = \frac{j}{\sigma} \Rightarrow E \propto \frac{1}{A}$$

$$E \propto \frac{1}{\pi(r_1 + ba)^2}$$

$$H = \frac{i^2 \rho}{A}$$

$$H \propto \frac{1}{A}$$

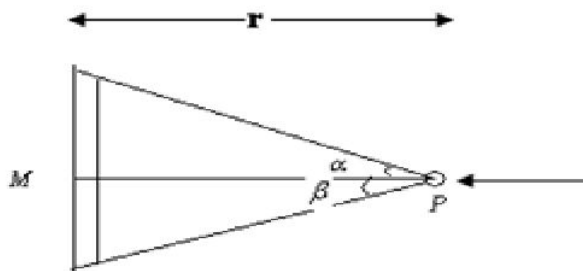
$$\frac{H}{E} = i, \text{ constant}$$

$$\int dv = \int_0^a \frac{\rho i dx}{\pi(r_1 + ab)^2}$$

$$v = \frac{\rho i}{\pi b} \left(\frac{ab}{r_1 + ab} \right)$$

43. Mass M is uniformly distributed over the rod of length L . The linear mass density of rod is λ

Then



A) If $\alpha = \beta$ then $I_y = 0$ ($I_y \rightarrow$ Gravitational intensity at P along y)

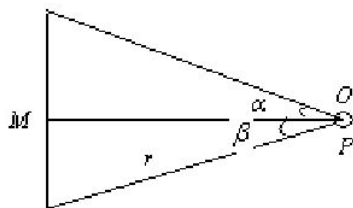
B) $I_x = \frac{2G\lambda}{r}$ if the length of the rod is infinite

C) Gravitational potential is not defined at P if the length of rod is infinite (By assuming potential at infinity is 0)

D) If $\alpha = \beta$ then $I_x = 0, I_y = 0$ (I_x, I_y are the gravitational intensities at P along x and y)

ABC

Sol. $I_x = \frac{G\lambda}{R} [\sin \alpha + \sin \beta]; I_y = \frac{G\lambda}{R} [\cos \alpha - \cos \beta]$



Now if

$$\text{Then } I_x = \frac{2G\lambda}{R} \sin \alpha; I_y = 0$$

So choice is (A) is correct and choice (D) is wrong

Relation between gravitational field intensity and gravitational potential

$$I = \frac{-dV}{dr}$$

$$\int_V^0 dV = - \int_r^\infty \frac{2G\lambda}{r} dr$$

$$[0 - V] = 2G\lambda [\log_e^\infty - \log_e^r]$$

$$V = 2G\lambda \log_e \left(\frac{r}{\infty} \right) \text{ (using definition of gravitational potential)}$$

$$\log_e \left(\frac{r}{\infty} \right) \text{ is undefined}$$

I.e, choice (C) is correct

44. Two different coils have self-inductances $L_1 = 8\text{mH}$ and $L_2 = 2\text{mH}$. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At that time, the current, the induced voltage and the energy stored in the first coil are i_1, V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are i_2, v_2 and w_2 respectively. Then

(a) $\frac{i_1}{i_2} = \frac{1}{4}$ (b) $\frac{i_1}{i_2} = 4$ (c) $\frac{W_1}{W_2} = \frac{1}{4}$ (d) $\frac{V_1}{V_2} = 4$

Ans: ACD

Sol: $P_1 = P_2 \Rightarrow V_1 i_1 = V_2 i_2$

$$\Rightarrow L_1 \frac{di_1}{dt} i_1 = L_2 \frac{di_2}{dt} i_2$$

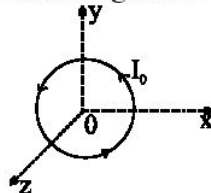
$$\Rightarrow \frac{i_1}{i_2} = \frac{L_2}{L_1} = \frac{1}{4}$$

$$\frac{V_1}{V_2} = \frac{L_1}{L_2} = \frac{4}{1}$$

$$\frac{W_1}{W_2} = \frac{L_1}{L_2} \left(\frac{i_1}{i_2} \right)^2 = 4 \left(\frac{1}{4} \right)^2 = \frac{1}{4}$$

45. A uniform ring of mass m and radius R carrying current I_0 is lying in the x - y plane in vacuum with centre at origin. A uniform external magnetic field of strength

$\vec{B} = B_0 (2\hat{i} - 2\hat{j} + 5\hat{k})$ Tesla is switched on at $t = 0$ (here $B_0 = \text{constant}$). Gravity is neglected. If the ring can freely rotate, then magnitude of



- A) Initial angular acceleration of ring is $\frac{4\sqrt{2}I_0B_0\pi}{m}$
 B) Initial angular acceleration of ring is $\frac{2\sqrt{2}I_0B_0\pi}{m}$
 C) The magnetic force acting on loop $2\sqrt{3}I_0\pi RB_0$
 D) The magnetic force acting on loop is zero

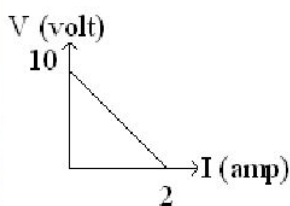
Ans: AD

Sol: $\vec{\psi} = (\vec{A} \times \vec{B})\hat{i}$

$$\vec{A} = (\pi R^2)\hat{k}$$

Magnetic force on a current loop is zero, when placed in a uniform magnetic field.

46. A battery of e.m.f E and internal resistance r is connected to resistance R . Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current passing through the resistance (I) and potential difference (V) across it. Select the correct alternatives.



- a) Internal resistance of the battery is 5Ω
 b) e.m.f of the battery is $10V$
 c) Maximum current which can be taken from the battery is $2A$
 d) $V-I$ graph can never be a straight line as shown in the Fig.

Ans: ABC

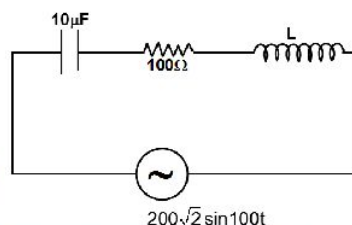
Sol: From the graph

$$E = V - ir \Rightarrow E = V = 10 \text{ when } i = 0$$

$$\text{When } i = 2 \Rightarrow E = 0 \Rightarrow r = 5\Omega$$

INTEGER/Numerical type Questions

47. A $10\mu\text{F}$ capacitor, an inductor and a resistor of $100\ \Omega$ are connected to an alternating source of emf $200\sqrt{2}\sin 100t$ as shown in the figure. What is the r.m.s current in the circuit and the value of the inductance if the current and the source voltage attain their maxima simultaneously? What is the average power consumed in this case in watt??



Ans: 400

Sol. The current and the voltage attain their maxima and minima simultaneously. Hence they are in phase and the circuit is in resonance.

$$Z = R = 100\ \Omega$$

$$I = \frac{V}{Z} = \frac{200\sqrt{2}}{100} = 2\sqrt{2}\text{ A}, I_{\text{r.m.s.}} = 2\text{ A}$$

$$\text{The power factor} = \frac{R}{Z} = 1$$

$$\text{Average power consumed} = VI \cos\phi$$

$$= 200 \times 2 \times 1 = 400 \text{ watt}$$

$$\text{and at resonance } \omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow 100 = \frac{1}{\sqrt{L \times 10 \times 10^{-6}}}$$

$$\Rightarrow L = \frac{1}{10^4 \times 10 \times 10^{-6}} = 10\text{ H}$$

48. A 200Km long telegraph wire has capacity of $0.014\mu\text{F/km}$. If it carries an alternating current of 50KHz, what should be the value of an inductance required to be connected in series so that impedance is minimum? (in Henry)

Ans: 0.303

Sol. $I_{\text{rms}} = \frac{90}{36} = \frac{5}{2} = 2.5 \text{ Amp}$

$$V_L = \sqrt{120^2 - 36^2} = \sqrt{156 \times 84} = 114.47 \text{ Volt}$$

$$\cos\phi = \frac{30}{120} = 0.3$$

$$\phi = 72.5^\circ$$

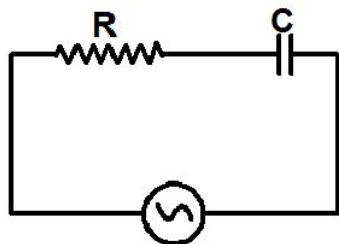
$$\tan 72.5^\circ = \frac{2\pi \times 60 \times L}{36} \Rightarrow L = 0.303 \text{ H}$$

48. An alternating current of 1.5mA and angular frequency $\omega = 300 \text{ rad/s}$ flows through $10\text{K}\Omega$

resistor and a $0.50\mu\text{F}$ capacitor in series. Find the r.m.s. voltage across the capacitor and impedance of the circuit?

Ans: 10

Sol: $i = 1.5 \text{ mA}$, $\omega = 300 \text{ rad/sec}$



$$\therefore X_C = \frac{1}{\omega C} = \frac{1}{300 \times 0.50 \times 10^{-6}} = \frac{10^5}{15} = 6.6 \times 10^3 \Omega$$

$$Z = \sqrt{R^2 + X_C^2} = \sqrt{(10000)^2 + (6600)^2} = 11.98 \text{ k}\Omega \approx 12 \text{ k}\Omega$$

$$\therefore V_C = iX_C = 1.5 \times 10^{-3} \times 6.6 \times 10^3 = 9.9 \approx 10 \text{ V}$$

49. When a 15 V dc source was applied across a choke coil then a current of 5 Amp flows in it. If the same coil is connected to a 15 V, 50 rad/s ac source a current of 3 Amp flows in the circuit. Determine the inductance of the coil. Also find the power developed in the circuit and its resonance frequency if a $2500 \mu\text{f}$ capacitor is connected in series with the coil. (in watt)

Ans: 27

Sol: For a coil,

$$Z = \sqrt{R^2 + \omega^2 L^2}$$

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\text{For dc source, } \omega = 0 \quad I = \frac{V}{R} \quad \text{i.e. } R = \frac{15}{5} = 3\Omega \quad \dots (i)$$

When ac is applied

$$I = \frac{V}{Z} \quad \text{i.e. } Z = \frac{15}{3.0} = 5\Omega$$

$$\therefore R^2 + X_L^2 = 25$$

$$X_L^2 = 25 - 9 = 16\Omega \Rightarrow X_L = 4\Omega$$

$$L = \frac{4}{40} = 0.08 \text{ Henry.}$$

Now when the capacitor is connected in series.

$$\therefore X_C = \frac{1}{\omega C} = \frac{1}{50 \times 2500 \times 10^{-6}} = 8\Omega$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{3^2 + (4 - 8)^2} = 5\Omega$$

$$\therefore I = \frac{15}{5} = 3\text{A}$$

$$\therefore P_{av} = V_{rms} I_{rms} \cos \phi = I_{rms}^2 R = (3)^2 \times 3 = 27 \text{ W.}$$

f Resonance frequency:-

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(0.08) \times 2500 \times 10^{-6}}} = \frac{1000}{2\pi \times 5 \times 2\sqrt{2}}$$

$$= \frac{25\sqrt{2}}{\pi} = 11.25 \text{ Hz.}$$

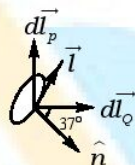
50. Radius of a very long solenoid consisting of 100 turns per meter, is 0.5 cm. Normal to the plane of each turn of the solenoid makes 37° with the axis of the solenoid. B on two concentric circles with their centres on the mid point of axis of the solenoid and with their planes perpendicular to the axis and of circumferences 1 cm and 6 cm are B_1 and B_2 respectively. $\frac{B_1}{B_2}$ is $\left(\cos 37^\circ = \frac{4}{5}\right)$

Ans : 8

Sol : B_{in} is due to the current perpendicular to the axis I_P and B_{out} is due to the current parallel to the axis I_Q

$$I|d\vec{l}_P| = I dl \cos 37^\circ = \frac{4Idl}{5}$$

$$I|d\vec{l}_S| = I dl \sin 37^\circ = \frac{3Idl}{5}$$

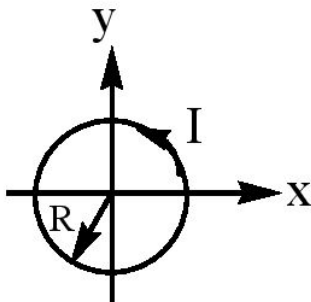


Since radius of solenoid is $R = 0.5\text{cm}$, circle of length 1 cm lies inside the solenoid and the circle of length 6cm lies outside of the solenoid. $\frac{B_1}{B_2} = \frac{B_{in}}{B_{out}} = \frac{\mu_0 n I_P}{\frac{\mu_0 I_Q}{2\pi r_2}} = n \frac{I_P}{I_Q} (2\pi r_2)$

Here r_2 is the radius of outer circle

$$\Rightarrow \frac{B_1}{B_2} = (100) \left(\frac{4}{3}\right) (6 \times 10^{-2}) = 8$$

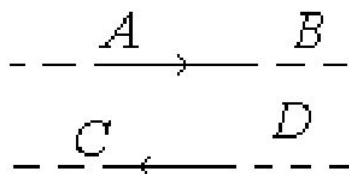
51. A circular coil of radius 3m is arranged in XY plane with its centre at the origin as shown in figure. If magnetic induction field strengths at the points $P(1, \sqrt{3}, \sqrt{2})$ and $Q(\sqrt{2}, \sqrt{2}, \sqrt{2})$ are B_P and B_Q respectively (all the coordinates are in m), then $\frac{B_P}{B_Q}$ is



Ans : 1

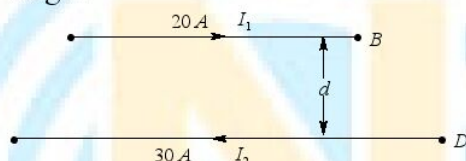
Sol. Both the points P and Q are at a distance 2 units from the origin in the XY plane and $\sqrt{2}$ units distance from the XY plane. Hence magnitudes of B must be same. $\frac{B_1}{B_2} = 1$

52. A long horizontal wire AB which is free to move in a vertical plane and carries a steady current of 20A is in equilibrium at a height of 0.02 m over another parallel long-wire CD which is fixed in a horizontal plane and carries a steady current of 30 A as shown in the figure. When AB is slightly depressed, it is found AB executes simple harmonic motion with a period $T = 2\pi\sqrt{\frac{x \times 10^{-2}}{9.8}}$ s. Find the value of x . [Take $g = 9.8 \text{ m/sec}^2$].



ANS:2

Sol: Consider force on unit length.



At equilibrium:

$$\therefore \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{d} = mg \quad m \rightarrow \text{mass per unit length.}$$

Let us depress AB by a small displacement 'y'.

$$\text{Now, unbalanced force, } F = \frac{\mu_0 I_1 I_2}{2\pi(d-y)} - mg = m\omega^2 y$$

$$\Rightarrow \frac{\mu_0 I_1 I_2}{2\pi d \left(1 - \frac{y}{d}\right)} - mg = m\omega^2 y \Rightarrow mg \left[1 - \frac{y}{d}\right]^{-1} - mg = m\omega^2 y$$

$$\Rightarrow mg \left[1 + \frac{y}{d}\right] - mg = m\omega^2 y \Rightarrow \omega = \sqrt{\frac{g}{d}}$$

$$T = 2\pi\sqrt{\frac{d}{g}}$$

$$\therefore T = 2\pi\sqrt{\frac{0.02}{9.8}}$$

$$\therefore T = 2\pi\sqrt{\frac{2 \times 10^{-2}}{9.8}}$$

53. Two long straight wires are located parallel to each other in space at a separation of $2b$. The wire cross-sectional radius is equal to a , find the mutual capacitance of this system per unit

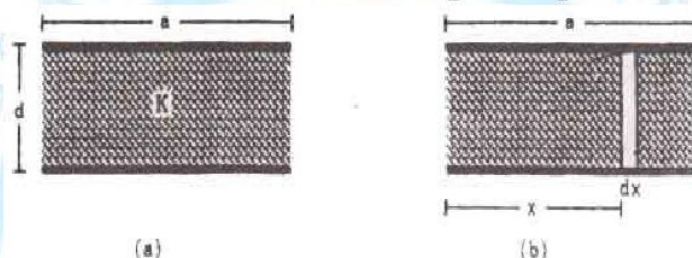
length of the wires under the condition $a \ll b$. $\frac{n\pi\epsilon_0}{\ln\left(\frac{2b}{a}\right)}$ Then n is

ANS:1

SOL. $E = \frac{\lambda}{2\pi\epsilon_0 x} + \frac{\lambda}{2\pi\epsilon_0 (b-x)} \quad \frac{1}{b-x}$

$$\frac{-dv}{dx} = \frac{\lambda}{2\pi\epsilon_0} \left[\frac{b}{x(b-x)} \right]$$

54. The figure shows a parallel-plate capacitor having square plates of edge a and plate-separation d . The gap between the plates is filled with a dielectric of dielectric constant k which varies as $K = K_0 + \alpha x$ parallel to an edge as where K and α are constants and x is the distance from the left end. Capacitance is $\frac{\epsilon_0 a^2}{d} \left[k_0 + \frac{\alpha a}{n} \right]$ n is



ANS:2

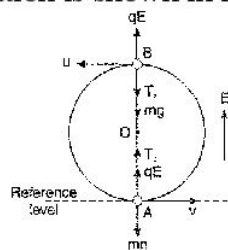
SOL. $dC = \frac{K(Ldx)\epsilon_0}{d}$ with $K = K_0 + \alpha x$

$$C = \int dC = \frac{\epsilon_0 L^2}{d} \left(K_0 + \frac{\alpha L}{2} \right)$$

55. A ball of mass m with a charge q can rotate in a vertical plane at the end of a string of length $l = 3.6m$ in a uniform electrostatic field whose lines of force are directed upwards. If horizontal velocity, $(k \times 4) \text{ ms}^{-1}$, must be imparted to the ball in the upper position so that the tension in the string in the lower position of the ball is 15 times than the weight of the ball, given $E = \frac{6mg}{5q}$ and $g = 10 \text{ ms}^{-2}$, find the value of ' k '.

ANS:6

SOL: The corresponding situation is shown in figure



Here by using work energy theorem between position B and A we have

$$\frac{1}{2}mu^2 + 2mgl - 2qEl = \frac{1}{2}mv^2$$

At position A we have

$$T + qE = \frac{mv^2}{l} + mg$$

Since $T = 15mg$

$$\Rightarrow qEl = mv^2 - 14mgl \quad \{\therefore \text{at the lowest point, } T = 15mg\}$$

From equation (1), we have

$$qEl + 14mgl = mu^2 - 4mgl - 4qEl$$

$$\Rightarrow u^2 = \frac{5qE}{m} + \frac{10mg}{m} \Rightarrow u = \sqrt{\frac{5qEl}{m} + 10gl} \Rightarrow u = \sqrt{\frac{5l}{m}(qE + 2mg)}$$

$$\Rightarrow u = \sqrt{\frac{5l}{m}\left(\frac{6mg}{5}\right) + 2mg} = \sqrt{\frac{5l}{m}\left(\frac{16mg}{5}\right)} \Rightarrow u = 4\sqrt{gl} = 4\sqrt{(10)(3.6)} = (4)(6) = 24ms^{-1}$$

56. A charged dust particle of radius $5 \times 10^{-7} m$ is located in a horizontal electric field having an intensity of $6.28 \times 10^5 Vm^{-1}$. The surrounding medium is air with coefficient of viscosity $\eta = 1.6 \times 10^{-5} Nsm^{-2}$. If this particle moves with a uniform horizontal speed $0.02 ms^{-1}$, find the number of electrons in dust particle in the form $(k \times 10)$ and give the value of 'k'

ANS:3

SOL. The horizontal force on dust particle is $F = qE$

If v is speed of particle in air, then by stoke's law,

Viscous force $= 6\pi\eta rv$

For uniform speed v , we have

$$qE = 6\pi\eta rv$$

If n is the number of excess electrons, then

$$q = ne$$

$$\Rightarrow neE = 6\pi\eta rv \Rightarrow n = \frac{6\pi\eta rv}{eE}$$

Substituting given values

$$n = \frac{6 \times 3.14 \times 1.6 \times 10^{-5} \times 5 \times 10^{-7} \times 0.02}{1.6 \times 10^{-19} \times 6.28 \times 10^5} \Rightarrow n = \frac{6 \times 3.14 \times 5 \times 2}{6.28} \Rightarrow n = 30$$

57. A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = kr^a$, where k and a are constants and r is the distance from its centre. If the electric field at $r = \frac{R}{2}$ is $\frac{1}{8}$ times that at $r = R$, Find the value of a .

Ans : 2

SOL. Total charge

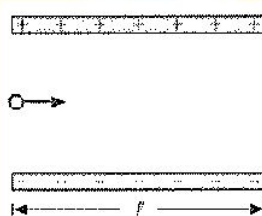
$$Q \int p dV = \int_{r=0}^{r=R} (Kr^a) (4\pi^2 dr) = \frac{4\pi K}{a+3} (R^{a+3})$$

$$Q' \int p dV = \int_{r=0}^{r=\frac{R}{2}} (Kr^a) (4\pi^2 dr) = \frac{4\pi K}{a+3} \left(\frac{R}{2}\right)^{a+3}$$

According to question $\frac{1}{4\pi\epsilon_0} \frac{Q'}{\left(\frac{R}{2}\right)^2} = \frac{1}{8} \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \right)$

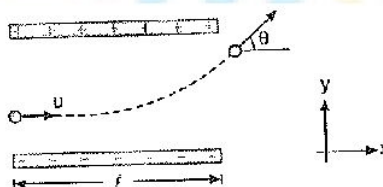
Putting the value of Q and Q' get $a = 2$

58. A uniform electric field E is created between two parallel charged plates as shown in figure. An electron enters the field symmetrically between the plates with a speed u . The length of each plate is l . If the angle of deviation of the path of the electron as it comes out of the field is $6k$ degrees for $u^2 = \frac{\sqrt{3}El e}{m}$ find k.



Ans : 5

SOL. A long the x-direction speed of electron remains uniform ; because of no component of field along horizontal. So,



$$V_x = u \text{ and } X = ut \Rightarrow l = ut$$

Along y direction

$$u_y = 0$$

Acceleration along y-direction of motion of electron is given by

$$a_y = a = \frac{eE}{m}$$

$$\text{Since } v_y = u_y + a_y t \Rightarrow v_y = 0 + \left(\frac{eE}{m} \right) t$$

$$\text{From (1), we get } t = \frac{l}{u}$$

$$v_{y \text{ final}} = v_{y \text{ initial}} + at \Rightarrow v_y = \left(\frac{eE}{m}\right)\left(\frac{l}{u}\right)$$

If θ be the angle of deviation, then

$$\tan \theta = \frac{v_y}{v_x} = \frac{\left(\frac{eEl}{mu}\right)}{u} \Rightarrow \theta = \tan^{-1}\left(\frac{eEl}{mu^2}\right)$$

$$\Rightarrow \theta = \tan^{-1}\left[\frac{eEl}{m\left(\sqrt{3}\frac{eEl}{m}\right)}\right] = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ$$

59. A particle of mass $2 \mu g$ and carrying a charge of $1 \times 10^{-9} C$ is moving directly towards a fixed positive point charge of magnitude $1 \times 10^{-8} C$. When it is at a distance of 10 cm from the fixed point charge it has a velocity of 1 ms^{-1} . If at $\left(\frac{94}{k}\right) \text{ mm}$ distance from the fixed point charge the particle comes momentarily to rest, find the value of ' k '

ANS:2

SOL. By Law of Conservation of Energy

$$\Rightarrow \frac{1}{2}mv^2 + \frac{Qq}{4\pi\epsilon_0 r_1} = \frac{1}{2}m(0)^2 + \frac{Qq}{4\pi\epsilon_0 r_2} \Rightarrow \frac{1}{2}mv^2 + \frac{Qq}{4\pi\epsilon_0} \left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

$$\Rightarrow (2 \times 10^{-6})(1)^2 = (10^{-9})(10^{-8})(9 \times 10^9) \left(\frac{1}{r_2} - \frac{1}{\left(\frac{10}{100}\right)}\right)$$

$$\Rightarrow \frac{100}{9} = \frac{1}{r_2} - 10 \Rightarrow \frac{1}{r_2} = \frac{190}{9} \Rightarrow r_2 = \frac{9}{190} \text{ m}$$

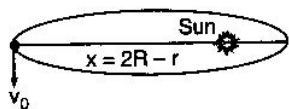
$$\Rightarrow r_2 = 4.7 \times 10^{-2} \text{ m} \Rightarrow r_2 = 47 \times 10^{-3} \text{ m} \Rightarrow r_2 = 47 \text{ mm}$$

60. A planet revolves about the sun in elliptical orbit of semi major axis $2 \times 10^{12} \text{ m}$. The areal velocity of the planet when it is nearest to the sun is $4.4 \times 10^{16} \text{ m}^2/\text{sec}$. The least distance between planet and the sun is $1.8 \times 10^{12} \text{ m}$. The minimum speed of the planet (in km/sec) is 10K. Determine the value of K.

Ans : 4

Sol. Area covered by line joining planet and sun in time dt is

$$ds = \frac{1}{2}x^2 d\theta$$



Areal velocity

$$= \frac{ds}{dt} = \frac{1}{2} x^2 \frac{d\theta}{dt} = \frac{1}{2} x^2 \omega$$

Where x = distance between planet and sun and ω = angular speed of planet about sun.

From Kepler's second law, areal velocity of planet is constant.

At farthest position

$$= \frac{ds}{dt} = \frac{1}{2} (2R - r)^2 \omega$$

$$= \frac{1}{2} (2R - r) (2R - r) \omega = \frac{1}{2} (2R - r) v_B$$

$$\text{or } v_B = \frac{2A}{2R - r} = \text{least speed.}$$

Using values, $v_B = 40 \text{ km/sec}$. thus $K = 4$.

61. A solid sphere of mass M and radius R is moving with speed $\sqrt{\frac{GM}{R}} = 100\sqrt{10} \text{ ms}^{-1}$ in free space. Due to application of external impulsive force, it breaks into eight spheres of same radius in such a way that the total mechanical energy before breaking and after breaking of sphere remains same. The speed of each sphere is.... m/s, when separation between them is very large. Assume at that instant, speed of each sphere is same. Give only first digit of your answer.

Ans : 1

Sol. Using Conservation of energy

$$-\frac{3GM^2}{5R} + \frac{1}{2} M v_0^2 = -\frac{3G(M/8)^2}{5(R/2)} \times 8 + \frac{1}{2} M v^2$$

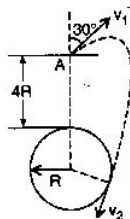
$$\text{or, } -\frac{3GM^2}{5R} + \frac{1}{2} M v_0^2 = -\frac{3GM^2}{20R} + \frac{1}{2} M v^2 \quad \text{or } \frac{GM}{20R} = \frac{v^2}{2}$$

$$\text{or } v = \frac{1}{\sqrt{10}} \times \sqrt{\frac{GM}{R}} = 100 \text{ m/sec}$$

62. A particle is projected from point A, that is at distance of $4R$ from the centre of the earth, with speed v_1 in a direction making angle of 30° with the line joining the centre of the earth and the point A, as shown. Find the speed v_1 , if particle passes grazing the surface of the earth.

Consider gravitational interaction only between these two points. Assume $v_1 = \sqrt{2}k$ (km/s).

The value of k is ?



Ans : 4

Sol. Conserving angular momentum

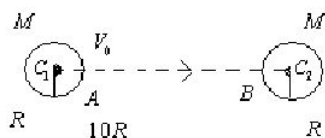
$$m(v_1 \cos 60^\circ)4R = mv_2 R \text{ or } \frac{v_2}{v_1} = 2$$

Conserving energy of the system

$$-\frac{GMm}{4R} + \frac{1}{2}mv_1^2 = -\frac{GMm}{R} + \frac{1}{2}mv_2^2 \text{ or } \frac{1}{2}v_2^2 - \frac{1}{2}v_1^2 = \frac{3}{4}\frac{GM}{R}$$

$$\text{or } v_1^2 = \frac{1}{2}\frac{GM}{R} \text{ or } v_1 = \frac{1}{\sqrt{2}} \times \sqrt{64 \times 10^6} = \frac{8000}{\sqrt{2}} \text{ m/s.}$$

63. Two large spherical objects of mass M each (uniformly distributed) are fixed as shown in the figure (distance between centers = $10R$). A small point mass m is projected from the surface of the first sphere from point A heading towards centre C_2 of second sphere. The minimum velocity of point mass so that it can reach upto second object at point B is $\frac{x}{3}\sqrt{\frac{GM}{5R}}$ then calculate $x =$



Ans : 8

Sol: the point mass has to cross at least the null point.

$$-\frac{GMm}{R} + \frac{-GMm}{9R} + \frac{1}{2}mv^2 = \frac{-2GMm}{5R}$$

64. The gravitational potential energy of a satellite revolving around the earth in circular orbit is -4MJ . Find the additional energy (in MJ) that should be given to the satellite so that it escapes from the gravitational force of the earth (consider only gravitational force on the satellite and no atmospheric resistance).

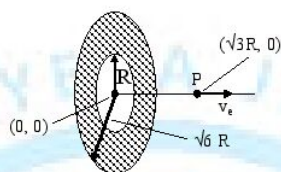
Ans : 2

Sol: $PE = -4MJ$

$TE = -2MJ$

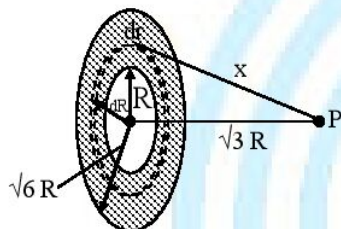
The additional energy required to make the satellite escape $= +2MJ$

65. A particle is projected from a point $P(\sqrt{3}R, 0)$ along the axis of an annular uniform disc of mass M and inner and outer radii R and $\sqrt{6}R$ respectively as shown in the figure. If the escape velocity of the particle from this point is $2\sqrt{\frac{GM}{N \times R}}$, find N



Ans: 5

Sol: Potential at point P due to elementary is equal to



$$dv = -\frac{Gdm}{x}$$

$$\text{where } dm = \frac{M}{\pi[(\sqrt{6}R)^2 - R^2]} 2\pi r dr = \frac{2}{5} M \frac{r dr}{R^2}$$

$$\text{Hence } dV = \frac{-G2Mrdr}{x5R^2}$$

$$\text{as } r^2 + (\sqrt{3}R)^2 = x^2$$

We can write, $2rdr = 2xdx$

$$\text{Hence } dV = \frac{-G2M}{5R^2x} x dx = \frac{-2}{5} \frac{GM}{R^2} dx$$

$$V = \int_{x_1}^{x_2} \frac{-2}{5} \frac{GM}{R^2} dx = -\frac{2}{5} \frac{GM}{R^2} (x_2 - x_1)$$

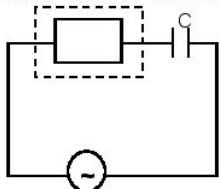
$$\text{Here, } x_2 = \sqrt{(\sqrt{3}R)^2 + (\sqrt{6}R)^2} = 3R$$

$$x_1 = \sqrt{(\sqrt{3}R)^2 + R^2} = 2R$$

Hence $V = -\frac{2}{5} \frac{GM}{R}$

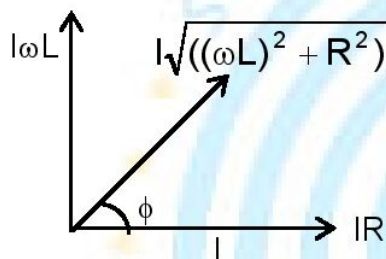
To escape a particle to infinity $\frac{1}{2}mv^2 \geq \frac{2}{5} \frac{GM}{R}m$

66. In the circuit shown there is a box and capacitance C connected to alternating power source of angular frequency of 2 rad/s . Box has power factor $1/\sqrt{2}$ and circuit has overall power factor 1. Find the admittance of the box in multiples of C .



Ans: 1.414

Sol: $\tan \phi = \frac{\omega L}{R}$



$1 = \frac{\omega L}{R} \quad (\cos \phi = 1/\sqrt{2}) \Rightarrow \tan \phi = 1$

$\omega L = R$

and overall circuit has power factor 1.

so, $\omega L = \frac{1}{\omega C}$

$Z = \sqrt{R^2 + (\omega L)^2} = \omega L \sqrt{2}$

$Z = \frac{1}{\omega C} \sqrt{2} = \frac{1}{C\sqrt{2}}$

67. An ac source of angular frequency ω is fed across resistor R and a capacitor C is series. The current registered is I . If now the frequency of source is changed to $\frac{\omega}{3}$ (but maintaining the same voltage), the current in the circuit is found to be halved. Calculate the ratio of reactance to resistance at the original frequency ω .

Ans: 0.77

Sol: $i = \frac{V}{\sqrt{R^2 + \left(\frac{1}{c\omega}\right)^2}} = \frac{V}{\sqrt{R^2 + X_c^2}}$

$$\frac{i}{2} = \frac{V}{\sqrt{R^2 + 9X_c^2}} = \frac{V}{2\sqrt{R^2 + X_c^2}} \Rightarrow 3R^2 = 5X_c^2$$

$$\Rightarrow \frac{X_c}{R} = \sqrt{\frac{5}{3}} = 0.77$$

68. A 750 Hz, 20 V source is connected to a resistance of 100Ω , an inductance of 0.1803 H and a capacitance of $10 \mu\text{F}$ all in series. Calculate the time in which the resistance (thermal capacity $2 \text{ J/}^\circ\text{C}$) will get heated by 10°C . (Ignore radiation) (in sec)

Ans: 350

Sol: $X_c = \frac{1}{CW} = \frac{1}{10 \times 10^{-6} \times 2 \times 750 \times \pi} = 21 \Omega$

$$X_L = LW = 0.1803 \times 2 \times 750 \times \pi = 850 \Omega$$

69. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude i each in the same direction, the radius of curvature of the path of the point charge is R_1 . In contrast, if the currents i in the two wires have directions opposite to each other, the radius of curvature of the path is R_2 . If $\frac{X_0}{X_1} = 3$,

the value of $\frac{R_1}{R_2}$ is

Ans : 3

Sol: $B_1 = \frac{\mu_0 i}{2\pi} \left[\frac{1}{x_1} - \frac{1}{(x - x_1)} \right]$

$$B_2 = \frac{\mu_0 i}{2\pi} \left[\frac{1}{x_1} + \frac{1}{x_0 - x_1} \right]$$

$$\text{or } R = \frac{mv}{Bq} \Rightarrow \frac{R_1}{R_2} = \frac{B_2}{B_1} = 1 - \frac{2x_1}{x_0}$$

70. Current in a toroid of rectangular cross section of 1000 turns is 2mA . Its inner and outer radii are 1m , and 4m respectively. If $\int \vec{B} \cdot d\vec{l}$ over a circular path of radius 2m with its centre on the axis of toroid and plane perpendicular to its axis and lies within the toroid is $\mu_0 x$, then x is

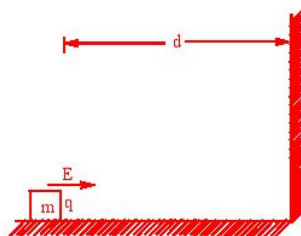
Ans : 2

Sol: According to Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_{\text{Net}}$$

$$i_{\text{Net}} - ni = 10^3 \times 2 \times 10^{-3} = 2$$

71. A block of mass m containing a net positive charge q is placed on a smooth horizontal table which terminates in a vertical wall as shown in figure. The distance of the block from the wall is d . A horizontal electric field 'E' towards right is switched on. Assuming elastic collisions (if any), if the time period of resulting oscillatory motion is $\sqrt{\frac{nmd}{qE}}$, find n .



ANS:8

Sol: From $s = ut + \frac{1}{2}at^2$

$$T = 2t = 2\sqrt{\frac{2d}{Eq/m}} = \sqrt{\frac{8md}{Eq}}$$

72. A satellite revolves round a planet in an elliptical orbit. Its maximum and minimum distance from the planet are 1.5×10^7 m and 0.5×10^7 m respectively. If the speed of the satellite at the farthest point be 5×10^3 m/s, the speed at the nearest point is 3a km/s. Determine the value of a.

Ans : 5

Sol: From conservation of angular momentum

$$mv_1r_1 = mv_2r_2$$

$$\Rightarrow v_2 = 15 \text{ kms}^{-1}$$

Matrix matching questions

73. A ring of mass 1 kg radius 2m carrying a current of 3A is placed in horizontal plane. It is free to rotate about a horizontal axis coinciding with its diameter.

A horizontal magnetic field of 4T is switched on such that it is perpendicular to the axis of rotation.

Match column I and II immediately after switching on magnetic field.

	Column I		Column II
(A)	Magnetic moment of ring	(p)	24π
(B)	Torque acting on ring	(q)	12π
(C)	Angular acceleration of ring	(r)	Zero
(D)	Net magnetic force on the ring	(s)	48π

$$A) (A-q); (B-s); (C-p); (D-r)$$

$$B) (A-q); (B-s); (C-r); (D-p)$$

$$C) (A-s); (B-r); (C-p); (D-q)$$

$$D) (A-s); (B-r); (C-q); (D-p)$$

Ans : A

Sol. $M = iA = i\pi r^2 = 3\pi (2)^2 = 12\pi$

$\tau = iAB = MB = 12\pi (4) = 48\pi$

$I\alpha = \tau \rightarrow \alpha = \frac{\tau}{\frac{1}{2}MR^2} = \frac{2 \times 48\pi}{1 \times 2^2} = 24\pi$

$w = \alpha t = \alpha (0) = 0$

74. Two spherical conductors of radii R_1 and R_2 have charges $+Q_1$ and $+Q_2$, capacitances C_1 and C_2 , surface charge densities σ_1 and σ_2 respectively. They are placed large distance apart and joined by a conducting wire. If Q'_1 and Q'_2 are the final charges and σ'_1 and σ'_2 are final charge densities and V_1 and V_2 are final potentials of the conductors then match the following:

	Column I		Column II
(A)	$\frac{C_1}{C_2}$	(p)	R_2^2 / R_1^2
(B)	$\frac{\sigma'_1}{\sigma'_2}$	(q)	1
(C)	$\frac{Q'_2}{Q'_1}$	(r)	$\frac{R_1}{R_2}$
(D)	$\frac{v_2}{v_1}$	(s)	$\frac{R_2}{R_1}$

A) $(A-r); (B-s); (C-p); (D-q)$

B) $(A-q); (B-p); (C-s); (D-r)$

C) $(A-r); (B-s); (C-s); (D-q)$

D) $(A-q); (B-s); (C-p); (D-r)$

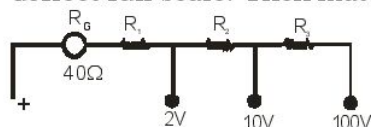
Ans : C

SOL. $C = 4\pi\epsilon_0 R$

$\sigma = \frac{Q}{4\pi R^2}$ $V = \frac{Q}{C} = \frac{Q}{4\pi\epsilon_0 R}$

$Q'_1 + Q'_2 = Q_1 + Q_2$ and $\frac{Q'_1}{C_1} = \frac{Q'_2}{C_2}$ $\therefore$ Final

75. Figure shows internal wiring of a three scale voltmeter whose binding parts are marked +2 V, 10 V and 100 V. When the meter is connected to the circuit being measured, one connection is made to the post marked + and the other to the post marked with the desired voltage range. The resistance of the moving coil R_G is 40Ω and a current of 1 mA in the coil causes it to deflect full scale. Then match the following :



Column A		Column B	
(A)	Value of R_1 in $K\Omega$	(p)	100
(B)	Value of R_3 in $K\Omega$	(q)	2
(C)	Overall resistance of the meter in 100 V range in $K\Omega$	(r)	1.96
(D)	Overall resistance of the meter in 2V range in $K\Omega$	(s)	90

ANS: (A – r)

(B – s)

(C – p)

(D – q)

SOL: For 2V range

$$I_G(R_G + R_1) = 2V \Rightarrow R_1 = \frac{2}{10^{-3}} - 40 = 1.96K\Omega$$

For 10 V range

$$I_G(R_G + R_1 + R_2) = 10V \Rightarrow R_2 = \frac{10}{1 \times 10^{-3}} - 40 - 1960 = 8K\Omega$$

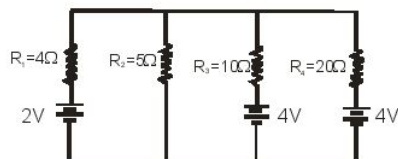
For 100 V range

$$I_G(R_G + R_1 + R_2 + R_3) = 100V \Rightarrow R_3 = \frac{100}{10^{-3}} - 40 - 1960 - 8000 = 90K\Omega$$

The overall resistance of the meter on 10 V range = $R_G + R_1 + R_2 = 40 + 1960 + 8000 = 10K\Omega$

The overall resistance of the meter on 2V range = $2K\Omega$

76. For the circuit given, match the power dissipated in resistors given in Column I with that in Column II. P_{R_1} , P_{R_2} , P_{R_3} and P_{R_4} are the power dissipated in resistors R_1 , R_2 , R_3 and R_4 respectively.

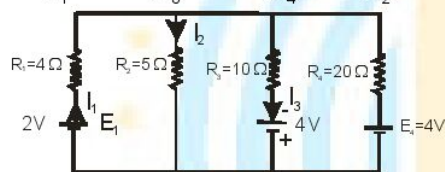


Column A		Column B	
(A)	P_{R_1}	(p)	49/80 Watt
(B)	P_{R_2}	(q)	2.025 Watt
(C)	P_{R_3}	(r)	9/16 Watt
(D)	P_{R_4}	(s)	0.05 Watt

Ans: A-R, B-S, C-Q, D-P

Sol: If voltage of terminal A is assumed to be V volts, then we may write Kirchoff's current law to that node as

$$\frac{E_1 - V}{R_1} + \frac{E_3 - V}{R_3} + \frac{E_4 - V}{R_4} = \frac{V}{R_2} \Rightarrow V = 0.5 \text{ Volts}$$



$$I_1 = \left(\frac{2 - 0.5}{4} \right) = \frac{3}{8} \text{ A}; \quad I_2 = \frac{0.5}{5} = 0.1 \text{ A}; \quad I_3 = \left(\frac{4 + 0.5}{10} \right) = 0.45 \text{ A}; \quad I_4 = \left(\frac{4 - 0.5}{20} \right) = \frac{7}{40} \text{ A}$$

$$P_{R_1} = \left(\frac{3}{8} \right)^2 \times 4 = \frac{9}{16} \text{ W}; \quad P_{R_2} = (0.1)^2 \times 5 = 0.05 \text{ W}; \quad P_{R_3} = (0.45)^2 \times 10 = 2.025 \text{ W}$$

$$P_{R_4} = \left(\frac{7}{40} \right)^2 \times 20 = \left(\frac{49}{80} \right) \text{ W}$$

77. Column A gives certain situations in which a straight metallic wire of resistance R is used and Column B gives some resulting effects. Match the statements in Column A with the statements in Column B and indicate your answer by darkening appropriate bubbles in the 4 x 4 matrix given in the ORS.

Column A		Column B	
(A)	A charge capacitor is connected to the ends of the wire	(p)	A constant current flows through the wire.
(B)	The wire is moved perpendicular to its length with a constant velocity in a uniform magnetic field perpendicular to the plane of motion	(q)	Thermal energy is generated in the wire
(C)	The wire is placed in a constant electric field that has a direction along the length of the wire	(r)	A constant potential difference develops between the ends of the wire
(D)	A battery of constant emf is connected to the ends of the wire.	(s)	Charges of constant magnitude appear at the ends of the wire

ANS.: (A – q), (B – r, s), (C – r), (D – p, q, r)

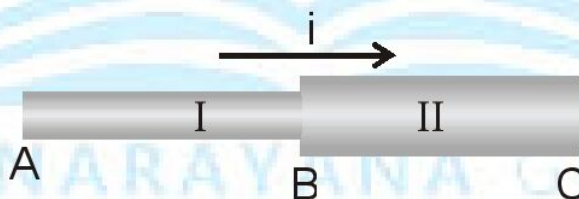
Sol. (A) Since wire will offer resistance, (A) → (q)

(B) Motional emf produced in a conductor of length l in a magnetic field B is Bv with the charges at the ends of length l . So, (B) → (r) and (s)

(A) Charges moved to the ends of the conductor in the field show potential difference. So, (C) → (r)

(B) Since on connecting a battery, potential difference is created, which allows current to flow and cause heat. So, (D) → (p), (q) and (r)

78. List I gives physical quantities of a situation in which a current i passes through two rods I and II of equal length that are joined in series. The ratio of free electron density (n), resistivity (r) and cross-section area (A) of both are in ratio $n_1 : n_2 = 2 : 1$, $r_1 : r_2 = 2 : 1$ and $A_1 : A_2 = 1 : 2$ respectively. List II gives corresponding results. Match the ratios in List I with the values in list II and select the correct answer using the code given below the lists



List I

- (P) $\frac{\text{Drift velocity of free electron in rod I}}{\text{Drift velocity of free electron in rod II}}$
- (Q) $\frac{\text{Electric field in rod I}}{\text{Electric field in rod II}}$
- (R) $\frac{\text{Potential difference across rod I}}{\text{Potential difference across rod II}}$
- (S) $\frac{\text{Average time taken by free electron to move from A to B}}{\text{Average time taken by free electron to move from B to C}}$

List II

- (1) 0.5
- (2) 1
- (3) 2
- (4) 4

(A) P-2, Q-4, R-4, S-2

(B) P-4, Q-2, R-1, S-2

(C) P-1, Q-3, R-2, S-2

(D) P-1, Q-3, R-1, S-3

Ans: A

SOL) (P) Since current in both rods is same.

$$\therefore n_1 e v_1 A_1 = n_2 e v_2 A_2$$

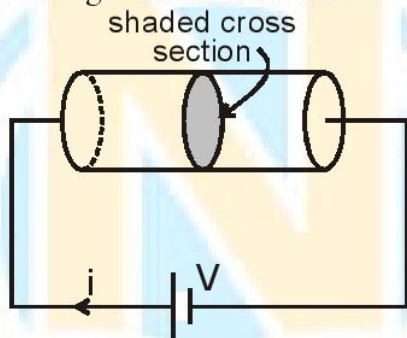
$$\therefore \frac{V_1}{V_2} = \frac{n_2 A_2}{n_1 A_1} = \frac{1}{2} \times \frac{2}{1} = 1$$

$$(Q) \ominus E = \rho J = \rho \frac{I}{A} \quad \therefore \frac{E_1}{E_2} = \frac{\rho_1}{\rho_2} \times \frac{A_2}{A_1} = \frac{2}{1} \times \frac{2}{1} = 4$$

$$(R) \frac{p.d. \text{ across rod I}}{p.d. \text{ across II}} = \frac{E_1 \times AB}{E_2 \times BC} = 4$$

$$(S) \frac{\text{Average time taken by free electron to move from A to B}}{\text{Average time taken by free electron to move from B to C}} = \frac{AB}{V_1} \times \frac{V_2}{BC} = 1$$

79. List I gives physical quantities based on a situation in which an ideal cell of emf V is connected across a cylindrical rod of uniform cross-section area and conductivity (σ) as shown in figure. E , J , ϕ and i are electric field at, current density through, electric flux through and current through shaded cross-section respectively as shown in figure. Physical quantities in list II are related to those in column I. Match the expressions in list I with the statements in list II and select the correct answer using the code given below the lists



List I

(P) $\frac{\phi}{i}$

(Q) $\frac{E}{J}$

(R) $\sigma \phi V$

(S) $\frac{V}{\sigma \phi}$

(A) P-1, Q-4, R-4, S-3

(C) P-2, Q-1, R-2, S-3

List-II

(1) Conductivity of rod

(2) Resistance of rod

(3) Resistivity of rod

(4) Power delivered to rod

(B) P-3, Q-3, R-4, S-2

(D) P-2, Q-3, R-2, S-1

Ans: B

SOL: From relation $J = \sigma E$

multiplying both sides by cross-section area of rod A we get

$$JA = \sigma EA \text{ or } i = \sigma \phi$$

$$\therefore \frac{\phi}{i} = \frac{1}{\sigma} = \text{resistivity of rod}$$

$$\frac{E}{J} = \frac{1}{\sigma} = \text{resistivity of rod}$$

$$\phi V = iV = \text{power delivered to rod}$$

$$\frac{V}{\sigma \phi} = \frac{V}{i} = \text{Resistance of rod.}$$

80. The following table gives the lengths of four copper rods at the same temperature, their diameters, and the potential differences between their ends.

Rod	Length	Diameter	Potential Difference
1	L	3d	V
2	2L	d	3V
3	3L	2d	2V
4	3L	d	V

Match the statements in List I with the current element in List II and select the correct answer using the code given below the lists

List I

- (P) Greatest Drift speed of the electrons.
 (Q) Greatest Current
 (R) Greatest rate of thermal energy produced
 (S) Greatest Electric field
 (A) P-4, Q-2, R-1, S-2
 (C) P-2, Q-4, R-2, S-2

List II

- (1) Rod 1
 (2) Rod 2
 (3) Rod 3
 (4) Rod 4

(B) P-3, Q-3, R-2, S-1

(D) P-2, Q-1, R-1, S-2

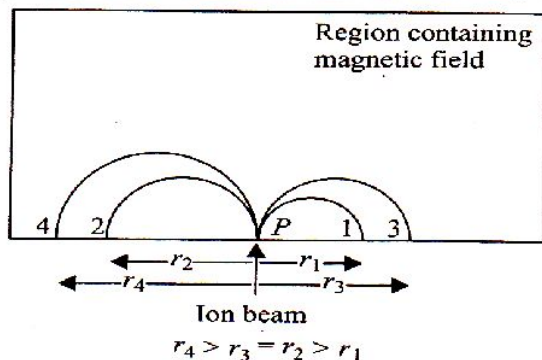
Ans: D

Sol: Drift speed $V_d = \frac{J}{ne} = \frac{i}{neA}$

$$i = \frac{V}{R} \text{ where } R = \frac{\rho L}{A}$$

$$E = \frac{V}{L} \text{ and } P = I^2 R$$

81. A beam consisting of four types of ions A, B, C and D enters a region at P that contains a uniform magnetic field as shown in fig. The field is perpendicular to the plane of the paper, but its precise direction is not given. All ions in the beam travel with the same speed. The table given below shows the masses and charges of the ions.



Ion
A

Mass
2m

Charge
+e

B	4m	-e
C	2m	-e
D	m	+e

The ions fall at different positions 1,2,3 and 4 as shown. Correctly match the ions with their respective positions of fall.

Column - I	Column - II
A) A	p)4
B) B	q)2
C) C	r)1
D) D	s)3

A) A-p,B-s,C-q,D-r

B) A-s,B-p,C-q,D-r

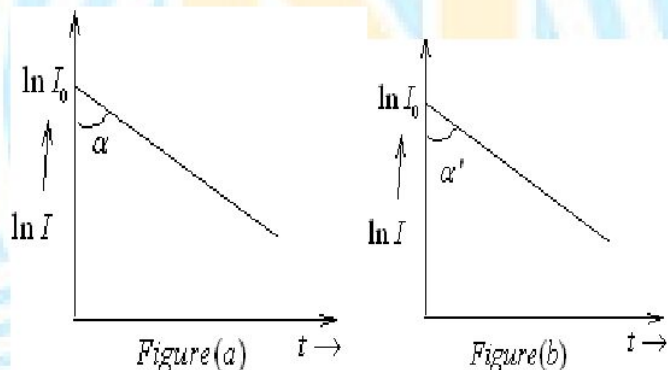
C) A-q,B-s,C-r,D-p

D) A-r,B-q,C-s,D-p

Ans : B

Sol: Force acting on a charged particle is given as $\vec{F} = q(\vec{v} \times \vec{B})$

82. I vs t curves for two series R-C circuits are given in figure (a) and figure (b). The capacitors are initially uncharged and battery is connected at $t=0$. Figure (a) corresponds to RC circuit with resistance and capacitance R and C respectively. Figure (b) corresponds to RC circuit with resistance and capacitance R' and C' respectively. Column-I gives various relations and Column-II gives possible values of R' and C' . Match respectively.



Column-I	Column-II
A) $\alpha' > \alpha$	P) $C' = C, R' = 2R$
B) $\alpha' < \alpha$	Q) $C' = 2C, R' = R$
C) $I'_0 > I_0$	R) $C' = \frac{C}{4}, R' = 2R$
D) $I'_0 < I_0$	S) $C' = 4C, R' = \frac{R}{2}$
	T) $C' = \frac{C}{8}, R' = 2R$

Ans: A-PQS ; B-RT ; C-S ; D-PRT

Sol: In an R-C circuit, $i = i_0 e^{-t/RC}$ or $\ln i = \ln i_0 - t/RC$

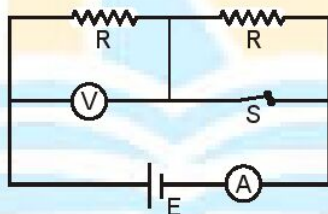
83. Match the following

Column I		Column II	
(A)	When a dielectric slab is slowly inserted between the plates of an isolated charged parallel plate capacitor	(p)	The electric potential energy of the system decreases
(B)	When a dielectric slab is slowly inserted between the plates of a charged parallel plate capacitor and its potential difference across the plates is kept constant	(q)	Work done by external agent is positive
(C)	When the plates of a charged parallel plate capacitor are pulled apart keeping its potential difference across the plates is kept constant	(r)	Work done by battery is positive
(D)	When the plates of a charged parallel plate capacitor are pulled apart, keeping charge constant.	(s)	Work done by external agent is negative.

Ans: A-PS ; B-RS ; C-PQ ; D-Q

Sol: $U_{\text{cap}} = \frac{q^2}{2C}$ for a charged and isolated capacitor

84. In the circuit shown, battery, ammeter and voltmeter are ideal and the switch S is initially closed as shown. What will be the changes in parameters as stated in list I if the switch S is opened. Possible changes are stated in list II. Match the expressions in list I with the statements in list II and select the correct answer using the code given below the lists



List I

- (P) Equivalent resistance across the battery
 (Q) Power dissipated by left resistance R
 (R) Voltmeter reading
 (S) Ammeter reading

List II

- (1) Remains same
 (2) Increases
 (3) decreases
 (4) Becomes zero.

(A) P-2, Q-3, R-3, S-3

(B) P-4, Q-2, R-1, S-3

(C) P-2, Q-1, R-1, S-3

(D) P-4, Q-2, R-3, S-3

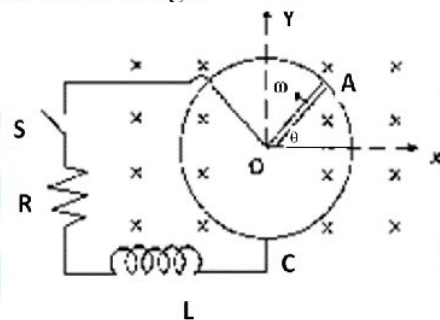
ANS: A

Sol: General application of Kirchhoff's laws.

COMPREHENSION TYPE

Paragraph:

A metal rod OA of mass m and length r is kept rotating counter clockwise with a constant angular speed ω in a vertical plane about a horizontal axis at the end O. The free end A is arranged to slide without friction along a fixed conducting circular ring in the same plane as that of rotation. A uniform and constant magnetic induction is applied perpendicular and into the plane of rotation as shown in figure. An inductor L and an external resistance R are connected through a switch S between the point O and a point C on the ring to form an electrical circuit. Neglect the resistance of the ring and the rod. Initially, the switch is open. S is closed at $t = 0$. Answer the following:



85. Expression for the current as a function of time is :

$$(A) I = \frac{B\omega r^2 [1 - e^{-Rt/L}]}{R}$$

$$(B) I = \frac{B\omega r^2 [1 - e^{-Rt/L}]}{2R}$$

$$(C) I = \frac{B\omega r^2 [1 - e^{-Rt/L}]}{4R}$$

$$(D) I = \frac{2B\omega r^2 [1 - e^{-Rt/L}]}{R}$$

Ans : B

Sol. $I = I_0 (1 - e^{-t/\tau})$, $I_0 = I(t \rightarrow \infty) = \frac{\mathcal{E}}{R}$ and $\tau = \frac{L}{R}$

86. The time dependence of the torque required to maintain the constant angular speed after steady state. Given that the rod OA was along the positive X-axis at $t = 0$ is :

$$(A) \tau = \frac{mgr}{2} \sin \omega t + \frac{\omega B^2 r^4}{2R}$$

$$(B) \tau = \frac{mgr}{2} \cos \omega t + \frac{\omega B^2 r^4}{2R}$$

$$(C) \tau = \frac{mgr}{2} \sin \omega t + \frac{\omega B^2 r^4}{4R}$$

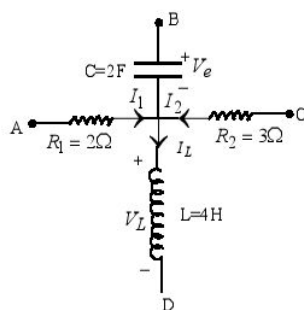
$$(D) \tau = \frac{mgr}{2} \cos \omega t + \frac{\omega B^2 r^4}{4R}$$

Ans : D

Sol. Restoring torque = $mg \frac{r}{2} \cos \omega t + \int_{r=0}^r B i d r r$

Paragraph :

In the figure shown, $I_1 = 10e^{-2t}$ A, $I_2 = 4$ A and $V_c = 3e^{-2t}$ V



87. The current I_L is :

- (a) $2e^{-2t}$ A (b) $2[2 - e^{-2t}]$ A (c) $[3 - 2(1 - e^{-2t})]$ A (d) $[2 + 3(1 - e^{-2t})]$ A

Ans : B

Sol. $I_C = \frac{dQ_C}{dt} = \frac{d}{dt}(6e^{-2t}) = -12e^{-2t}$

$$I_1 + I_2 + I_C = I_L$$

88. The potential difference across the inductor V_L is :

- (a) $8e^{-2t}$ v (b) $9e^{-2t}$ v (c) $16e^{-2t}$ v
(d) $18e^{-2t}$ v

Ans : C

Sol. $V_L = L \frac{dI_L}{dt}$

PARAGRAPH

A particle of mass m and charge q is accelerated by a potential difference V volt and made to enter a magnetic field region at an angle θ with the field. At the same moment, another particle of same mass and charge is projected in the direction of the field from the same point. Magnetic field of induction is B .

89. What would be the speed of second particle so that both particles meet again and again after a regular interval of time, which should be minimum?

- A) $\sqrt{\frac{qV}{m}} \cos \theta$ B) $\sqrt{\frac{2qV}{m}} \cos \theta$ C) $\sqrt{\frac{qV}{m}} \sin \theta$ D) $\sqrt{\frac{qV}{2m}} \cos \theta$

Ans : B

Sol. their velocities along the axis must be the same. They meet at pitch

90. Find the time interval after which they meet for first time

- A) $\frac{2\pi m}{qB}$ B) $\frac{\pi m}{2qB}$ C) $\frac{\pi m}{qB}$ D) $\frac{3\pi m}{2qB}$

Ans : A

Sol. Their velocities along the axis must be the same.

Paragraph ;

A charged particle carrying charge $q = 10 \mu C$ moves with velocity $v_1 = 10^6 \text{ ms}^{-1}$ at angle 45° with x-axis in the xy plane and experiences a force $F_1 = 5\sqrt{2}$ milli Newton along the negative z-axis. When the same particle moves with velocity $v_2 = 10^6 \text{ ms}^{-1}$ along the z-axis, it experiences a force F_2 in y-direction

91. Find the magnetic field $\vec{B}$

- A) $(10^{-3} T)(\hat{i} + \hat{j})$ B) $(2 \times 10^{-3} T)\hat{i}$ C) $(10^{-3} T)\hat{i}$ D) $(2 \times 10^{-3} T)(\hat{i} + \hat{j})$

Ans : C

Sol. Let $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$

$$\text{Apply } \vec{F} = q(\vec{v} \times \vec{B})$$

92. Find the magnitude of the force F_2

- A) $10^{-2} N$ B) $10^{-3} N$ C) $10^{-4} N$ D) $10^{-5} N$

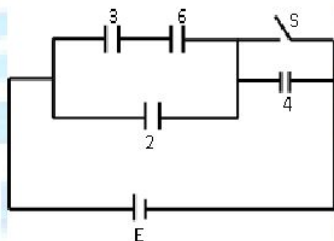
Ans : A

Sol. Let $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$

$$\text{Apply } \vec{F} = q(\vec{v} \times \vec{B})$$

Passage

In the circuit shown, the capacitances of the capacitors are shown in digits in farads. Each capacitor can withstand $10|C|$ volts at maximum where C is capacitance of that capacitor.



93. With switch S open, maximum energy that can be stored in the circuit is

- A) 2000 J B) 1600 J C) 3600 J D) None of these

Ans: B

Sol. Taking E as emf calculate charge on each capacitor.

$$Q \text{ sent by cell} = 2E$$

For each capacitor use $\frac{q}{c} < 10|c|$ to find E value

Of all Es found thus, $E_{\max}$ is less of these values.

$$u = \frac{1}{2} c E_{\max}^2$$

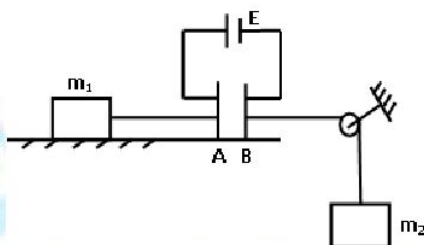
94. After closing the switch, charge flown through it is ($E = 10 \text{ V}$)
 A) 20 C B) 40 C C) 60 C D) None of these

Ans: B

Sol. Find charges on capacitors with switch in open and then in closed condition.
 Use charge conservation

Paragraph

The system shows a capacitor with smooth plates A and B lying on a non conducting surface. (Plates of capacitor are free to move horizontally) Coefficient of friction between m_1 and surface is μ . System is in equilibrium.



95. If $\mu < 1$ and m_2 is 2 kg, value of m_1 can't be
 A) 1.5 Kg B) 2.5 Kg C) 4 Kg D) None of these

Ans: A

Sol. Tension in string = force between plates

$$T = \frac{Q^2}{2A\epsilon_0} = m_2 g \quad \dots\dots\dots (1)$$

$$\text{And } T \leq \mu m_1 g \quad \dots\dots\dots (2)$$

96. If $m_1 = m_2$ and ($\mu < 1$). If system is initially in equilibrium and E is gradually increased.
 A) m_1 moves first B) m_2 moves first
 C) m_1 and m_2 move simultaneously D) Data is sufficient

Ans: A

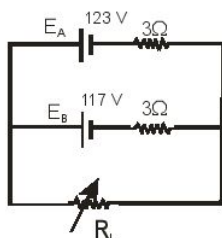
Sol. System in equilibrium $\Rightarrow m_2$ is balanced $m_2 g = T = \frac{Q^2}{2A\epsilon_0} \uparrow$ and hence m_2 moves

Simultaneously m_1 moves if $T = \mu m_1 g$

Simultaneously m_1 doesn't move if $T < \mu m_1 g$

Paragraph:

A battery E_A of 123 volts emf and internal resistance 0.4Ω and a battery E_B of 117 V emf and internal resistance 0.6Ω supplies a current to load resistance R_L through four wires of resistors $R_A = 1.2\Omega$, $R_B = 1\Omega$, $R_C = 1.4\Omega$ and $R_D = 1.4\Omega$ as shown in figure



97. The value of load resistance for maximum load power is
 (A) 1.5Ω (B) 3Ω
 (C) 4.5Ω (D) 6Ω

Ans : A

Sol: The network is redrawn

$$E_{eq} = \frac{E_A r_B + E_B r_A}{r_A + r_B} = 120 \text{ Volts} \quad R_{eq} = \frac{r_A r_B}{r_A + r_B} = 1.5\Omega$$

Hence, load resistance for max load power is $R_L = 1.5\Omega$

98. The maximum load power is
 (A) 800 W (B) 1600 W
 (C) 2400 W (D) 3200 W

Ans : C

Sol: Load current $I_L = \frac{E_{eq}}{R_{eq}} = \frac{120V}{3\Omega} = 40A$

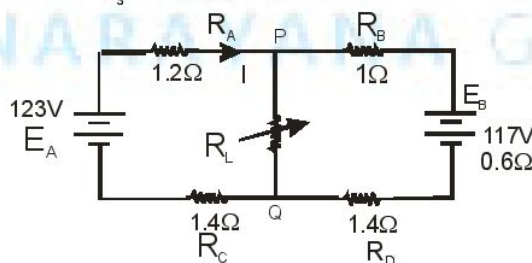
Maximum load power $P_L = I_L^2 R_L = (40)^2 (1.5) = 2400W$

99. The overall efficiency of the network is
 (A) 25 % (B) 50 %
 (C) 75 % (D) 100 %

ANS: B

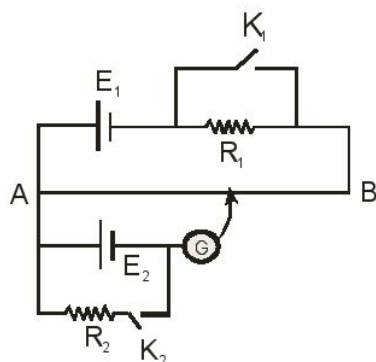
Sol. Power supplied by the source $P_s = E_{eq} I_L = 120 \times 40 = 4800W$

Efficiency of the network $\eta = \frac{P_L}{P_s} = \frac{2400}{4800} \times 100 = 50\%$



Paragraph:

In the circuit shown, AB is a 10Ω uniform slide wire 50 cm long. E_1 is 2V accumulator of negligible internal resistance. R_1 and R_2 are 15Ω and 5Ω respectively. When K_1 and K_2 are both open, the galvanometer shows no deflection when $AJ = 31.25$ cm when K_1 and K_2 are both closed the balance length $AJ = 5$ cm



100. The emf of the cell E_2 ,

- (A) 0.5V (B) 1V
(C) 1.5V (D) 2V

Ans : A

Sol. K_1 is open

$$\text{Potential gradient } K = \frac{2}{125} \times 10 \times \frac{1}{50} \text{ V/cm} = \frac{2}{125} \text{ V/cm}$$

$$E_2 = \frac{2}{125} \times 31.25 = 0.5 \text{ Volt}$$

101. The internal resistance of the cell E_2 ,

- (A) 7.5Ω (B) 8Ω
(C) 10Ω (D) 2Ω

Ans : A

Sol : K_1 is closed .

$$\text{Potential gradient } K' = \frac{2}{50} = \frac{1}{25} \text{ V/cm ;}$$

$$E_2 = K'L_1 \Rightarrow L_1 = 12.5\text{cm}; L_2 = 5\text{cm}$$

$$r_2 = R \left(\frac{L_1}{L_2} - 1 \right) = 5 \left(\frac{12.5}{5} - 1 \right) = 7.5\Omega$$

102. The balance length AJ when K_2 is closed and K_1 is open

- (A) 12.5 cm (B) 13.5 cm
(C) 14.5 cm (D) 15.5 cm

Ans : A

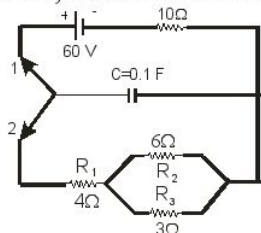
Sol. When K_2 is closed and K_1 is open, P.D across secondary circuit is,

$$V = \frac{E_2}{r + R_2} \times R_2 = \frac{0.5}{7.5 + 5} \times 5 = \frac{1}{5} = P\ell$$

$$\Rightarrow \ell = 12.5\text{cm}$$

Paragraph:

A two way switch S is used in the circuit as shown in figure. First the capacitor is charged by putting the switch in position 1. Now, switch is thrown to position 2.



103. Just before the switch is thrown to position 2, the energy stored in the capacitor is :

- (A) 60 J (B) 180 J
(C) 120 J (D) 200 J.

Ans : B

Sol : Energy stored in the capacitor $U = \frac{1}{2}CV^2 = \frac{1}{2}(0.1)(60)^2 = 180 \text{ J}$

104. Ratio of the heat energy dissipated in R_1, R_2 and R_3 is

- (A) 3 : 1 : 2 (B) 4 : 1 : 2
(C) 6 : 1 : 2 (D) 4 : 1 : 3

Ans : C

Sol. When switch is thrown to position 2 the capacitor starts discharging. Let i_1, i_2, i_3 be the current through R_1, R_2 and R_3 respectively at any instant.

$$\text{Then, } i_1 = i_2 + i_3 \dots \dots \dots (i)$$

$$6i_2 = 3i_3 \dots \dots \dots (ii)$$

Thus, the ratio of currents

$$i_1 : i_2 : i_3 = 3 : 1 : 2$$

The ratio remains constant throughout

Therefore, ratio of heat energy dissipated in R_1, R_2 and R_3 is :

$$H_1 : H_2 : H_3 = (3^2) \times 4 : (1^2) \times 6 : (2^2) \times 3$$

$$6 : 1 : 2$$

105. Heat energy dissipated in 4Ω resistor is

- (A) 60 J (B) 80 J
(C) 120 J (D) 100 J

Ans : C

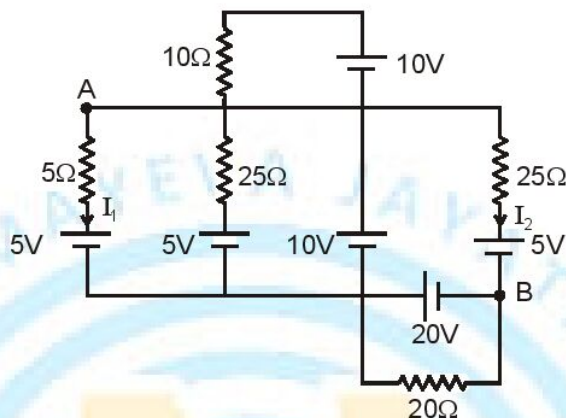
Sol. $H_1 : H_2 : H_3 = 6 : 1 : 2$

$$(H_1 + H_2 + H_3) = 180 \text{ J}$$

$$\Rightarrow H_1 = \frac{180}{9} \times 6 = 120 \text{ J} \text{ Hence, (C) option is correct.}$$

Paragraph :

The circuit consists of resistors and ideal cells. I_1 and I_2 are current through branches indicated in the figure. V_A and V_B is the potential at points A and B on the circuit.



106. The value of $\frac{I_2}{I_1}$ is :

- (A) 1 (B) 2 (C) 3 (D) 4

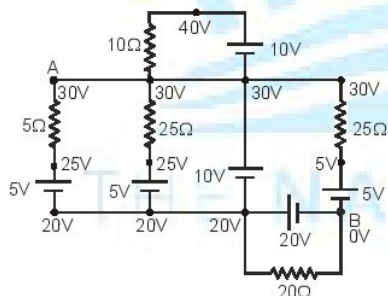
Ans : A

107. The value of $V_A - V_B$ in volts is :

- (A) 5 (B) 10 (C) 15 (D) 30

Ans : D

Sol: Assume the potential at B to be zero. The potential are given at various points in the circuit.



$$\text{Then } I_1 = \frac{30-25}{5} = 1\text{A and } I_2 = \frac{30-5}{25} = 1\text{A. Therefore } \frac{I_1}{I_2} = 1$$

Now from figure $V_A = 30\text{V}$ and $V_B = 0\text{V}$. Therefore $V_A - V_B = 30\text{V}$

Paragraph :

Resistance value of an unknown resistor is calculated using the formula $R = \frac{V}{I}$

where V and I be the readings of the voltmeter and the ammeter respectively. Consider the circuits below. The internal resistances of the voltmeter and the

ammeter (R_V and R_G respectively) are finite and non zero.

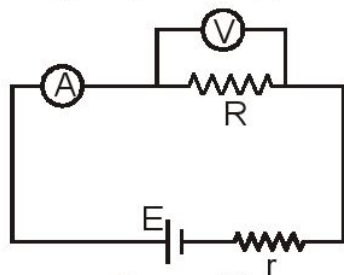


Figure (A)

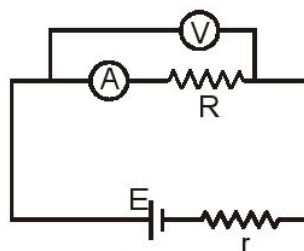
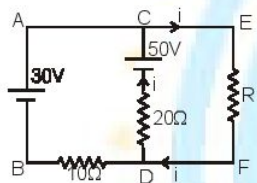


Figure (B)

Let R_A and R_B be the calculated values in the two cases A and B respectively.

108. The relation between R_A and the actual value R is
 (A) $R > R_A$ (B) $R < R_A$
 (C) $R = R_A$ (D) dependent upon E and r .
109. If the resistance of voltmeter is $R_V = 1k\Omega$ and that of ammeter is $R_G = 1\Omega$, the magnitude of the percentage error in the measurement of R (the value of R is nearly 100Ω) is :
 (A) zero in both cases (B) non zero but equal in both cases
 (C) more in circuit A (D) more in circuit B

SOL:

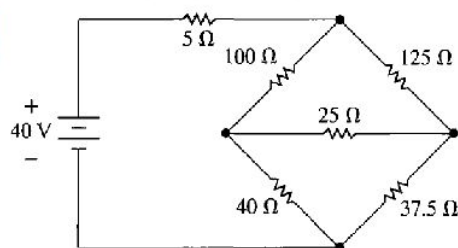


$$i = \frac{50}{20 + R} \quad \text{Potential drop across } R = \text{Potential drop across } AB$$

$$\Rightarrow \frac{50}{20 + R} R = 30 \Rightarrow R = 30\Omega$$

Paragraph :

The circuit shown below is a simple resistive circuit.



110. What is the power dissipated in the 5Ω resistance?
 (A) $\frac{5}{4} W$ (B) $5W$ (C) $15W$ (D) $\frac{1}{8} W$

Ans : A

Sol: The circuit is unbalanced Wheatstone bridge. Hence, by using Kirchhoff's laws, find the current in

each resistance can be obtained. By using $P=i^2R$, we can solve for power

111. If the $5\ \Omega$ resistor is to be replaced with a new resistor R , what should be the value of R for maximum power to be delivered to the R resistor.

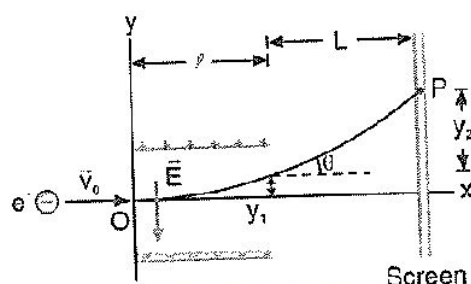
(A) $25\ \Omega$ (B) $5\ \Omega$ (C) $75\ \Omega$ (D) $80\ \Omega$

Ans : C

Sol: The circuit is unbalanced Wheatstone bridge. Hence, by using Kirchhoff's laws, the current in each resistance can be obtained. By using $P=i^2R$, we can solve for power

PARAGRAPH:

An electron is injected (at $t=0$) horizontally into a uniform electric field produced by two oppositely charged plates having length l and separation d , as shown in figure. The particle has an initial velocity $\vec{v}_0 = v_0 \hat{i}$ perpendicular to $\vec{E} = E \hat{j}$.



Charge is entering perpendicular to the electric field

Based on the above facts, and the figure provided, answer the following questions,

112. The electron makes an angle θ with the horizontal, when leaving the plates at time t_1 , then

(A) $\sin \theta = \frac{eEl}{mv_0^2}$ (B) $\sin \theta = \frac{eEl}{2mv_0^2}$ (C) $\tan \theta = \frac{eEl}{mv_0^2}$ (D) $\tan \theta = \frac{eEl}{2mv_0^2}$

Ans : C

Sol: when the electron leaves the plates at time t_1 , the electron makes an angle θ with the

$$\text{horizontal given by } \tan \theta = \frac{V_y}{V_x} = \frac{\frac{eE}{m} t_1}{v_0} = \frac{eEl}{mv_0^2}$$

113. The electron hits the screen located a distance L from the end of the plates at a time t_2 . The total vertical displacement of the electron from time $t=0$ until it hits the screen at t_2 is y . Then

(A) $y = \frac{eEl}{mv_0^2} \left(\frac{L}{2} + l \right)$ (B) $y = \frac{eEl}{mv_0^2} \left(\frac{l}{2} + L \right)$ (C) $y = \frac{eEl}{2mv_0^2} (L+l)$ (D) $y = \frac{eEl}{2mv_0^2} \left(L + \frac{l}{2} \right)$

Ans : B

Sol: After the electron leaves the plate, there is no longer any force on the electron so it travels in a

straight path. The deflection Y_2 is $Y_2 = L \tan \theta = \frac{eElL}{mv_0^2}$

$$\text{And the total deflection becomes } y = y_1 + y_2 = \frac{eEl^2}{2mv_0^2} + \frac{eElL}{mv_0^2} = \frac{eEl}{mv_0^2} \left(\frac{l}{2} + L \right)$$

Passage :

A particle of mass m_2 carrying a charge Q_2 is fixed on the surface of Earth. Another particle, mass m_1 and charge Q_1 , is positioned right above the first one at an altitude $h \ll R$, with R the radius of the Earth. The charges Q_1 and Q_2 have the same sign. Based on the above facts, answer the following questions.

114. The velocity of m_1 at a point p very close to Q_2 , a distance h_1 from the surface of earth, if the initial velocity of m_1 was zero and air drag and earth's electric field being ignored, is $[h_1 \ll h]$

(A) $\sqrt{2gh - \frac{Q_1 Q_2}{2\pi\epsilon_0 h_1 m_1}}$ (B) $\sqrt{2gh - \frac{Q_1 Q_2}{4\pi\epsilon_0 h_1 m_1}}$ (C) $\sqrt{2gh}$ (D) None of these

Ans : A

SOL. The forces of interaction, the force of gravity and Coulomb force are conservative, so we can apply the Law of conservation of Energy. We have

$$GPE_i + EPE_i = GPE_f + KE_f + EPE_f$$

$$m_1 gh + \frac{Q_1 Q_2}{4\pi\epsilon_0 h} = m_1 gh_1 + \frac{m_1 v^2}{2} + \frac{Q_1 Q_2}{4\pi\epsilon_0 h_1}$$

On solving for v , we get $v = \sqrt{2g(h - h_1) - \frac{2Q_1 Q_2(h - h_1)}{4\pi\epsilon_0 m h h_1}}$

Assuming that $h_1 \ll h$, equation (1) reduces to

$$v = \sqrt{2gh - \frac{2Q_1 Q_2}{4\pi\epsilon_0 m h_1}} = \sqrt{2gh - \frac{Q_1 Q_2}{2\pi\epsilon_0 m h_1}}$$

115. The magnitude of charge Q_2 at which the velocity of m_1 , at altitude h_2 is zero is given by

(A) $Q_2 = \frac{\pi\epsilon_0 m_1 g h h_2}{Q_1}$ (B) $Q_2 = \frac{4\pi\epsilon_0 m_1 g h h_2}{Q_1}$

(C) $Q_2 = \frac{2\pi\epsilon_0 m_1 g h h_2}{Q_1}$ (D) $Q_2 = \frac{\pi\epsilon_0 m_1 g h h_2}{4Q_1}$

Ans : B

SOL: Substituting $v = 0$ replacing h_1 by h_2 , we get

$$Q_2 = \frac{4\pi\epsilon_0 g m_1 h h_2}{Q_1}$$

116. At what altitude h_3 can the object m_1 be in equilibrium and what will be the nature of the object's oscillations if it is disturbed from equilibrium?

(A) $h_3 = \sqrt{\frac{Q_1 Q_2}{4\pi\epsilon_0 m_1 g}}$, periodic (B) $h_3 = \sqrt{\frac{Q_1 Q_2}{2\pi\epsilon_0 m_1 g}}$, periodic and oscillatory

(C) $h_3 = \sqrt{\frac{Q_1 Q_2}{4\pi\epsilon_0 m_1 g}}$, periodic and oscillatory (D) None of these

Ans : C

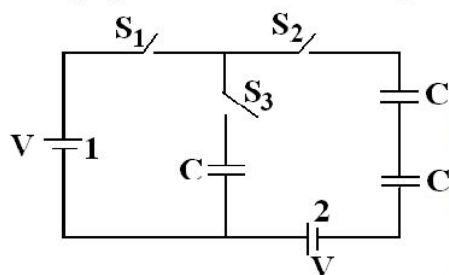
SOL. The object m_1 will be in equilibrium where gravitational and coulombic force balance,

$$\text{i.e., } \frac{Q_1 Q_2}{4\pi\epsilon_0 h_3^2} = m_1 g \Rightarrow h_3 = \sqrt{\frac{Q_1 Q_2}{4\pi\epsilon_0 m_1 g}}$$

$$a = g \sin 30^\circ - \mu g \cos 30^\circ - \frac{\mu q E}{m} \cos 60^\circ - \frac{q E}{m} \cos 30^\circ$$

Passage :

Three identical capacitors and two identical batteries of negligible internal resistance are connected as shown in figure. Neglect the resistance of the wires and all the three switches are initially open. There is no charge on any capacitor initially



117. S_2 and S_3 are closed and after some time S_1 is also closed. What is work done by battery 1 due to closing of switch S_1 ?

A) CV^2 B) $\frac{2}{3}CV^2$ C) $\frac{4}{3}CV^2$ D) $2CV^2$

Ans : D

118. In the above situation, what is work done by battery 2 due to closing of switch S_1 ?

A) Zero B) $-\frac{2}{3}CV^2$ C) $\frac{2}{3}CV^2$ D) $\frac{CV^2}{3}$

Ans: C

119. All the switches are closed for long time. Total energy stored in all the capacitors is U_1 . Now S_1 is open and total energy stored in all the capacitors is U_2 . What is the value of $U_1 - U_2$?

A) Zero B) $\frac{4}{3}CV^2$ C) $\frac{7}{12}CV^2$ D) $-\frac{4}{3}CV^2$

Ans : A

SOL:

When S_1 is open, charge on each capacitor is $\frac{CV}{3}$

When S_1 is closed, charge on each capacitor is CV

Charge sent by battery 2 is $\frac{2CV}{3}$

$$\therefore \text{W.D by it is } \frac{2CV^2}{3}$$

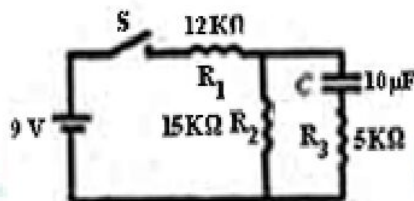
Charge sent by battery 1 is $2CV$

$$\therefore \text{W.D by it is } 2CV^2$$

When S_1 is again open, charges on the capacitors remain same

Passage :

Suppose the switch S has been closed sufficiently long for the capacitor to become fully charged, Find



120. The steady-state current through each resistor

A) $I_1 = I_2 = 3\text{mA}, I_3 = 0$

B) $I_1 = 1\text{A}, I_2 = 2\text{A}, I_3 = 3\text{A}$

C) $I_1 = I_2 = \frac{1}{3}\text{mA}, I_3 = 0$

D) $I_1 = I_2 = \frac{1}{3}\text{mA}, I_3 = 3\text{A}$

Ans : C

121. In the above question the switch S is now opened at $t=0$. Find the time that it takes for the charge on the capacitor to fall to half its initial value

A) 0.3 s

B) 0.1386 s

C) 0.693 s

D) 0.367 s

Ans : B

122. In the above question, write an expression for the current through R_2 as a function of time “s” open

A) $I = 0.25 \times e^{-5t} \text{mA}$

B) $I = 0.45 \times e^{-5t} \text{mA}$

C) $I = 0.25 \times e^{-10t} \text{mA}$

D) $I = 0.45 \times e^{-2t} \text{mA}$

Ans : B

$$\text{SOL: } U_{\text{cell}} = V(\Delta Q) = V \left[Q + \frac{Q_a}{cq - a} \right] = \frac{VQ^2}{Q - a}$$

$$q = \frac{q}{12+15} \text{mA}, I_3 = 0$$

Paragraph :

Two solenoids A and B spaced close to each other and sharing the axis have 400 and 700 turns respectively. A current of 3.5 A in coil A produced an average flux of $300\mu\text{T} - \text{m}^2$ through each turn of A and a flux of $90\mu\text{T} - \text{m}^2$ through each turn of B.

123. Calculate the self inductance of A.

(a) $1.23 \times 10^{-2} \text{H}$

(b) $2.34 \times 10^{-2} \text{H}$

(c) $3.43 \times 10^{-2} \text{H}$

(d) $4.53 \times 10^{-2} \text{H}$

Ans : C

Sol:

124. Calculate the mutual inductance of the two solenoids

(a) $0.5 \times 10^{-2} \text{H}$

(b) $3.4 \times 10^{-2} \text{H}$

(c) $4.6 \times 10^{-2} \text{H}$

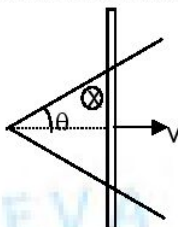
(d) $1.8 \times 10^{-2} \text{H}$

Ans : D

SOL:

SUBJECTIVE TYPE

125. A conducting bar of sufficient length is pulled with a constant velocity in a conducting $\wedge$ shaped rail as shown in the figure. Inward magnetic field of induction B is present inside the area bounded by the bar & the rail. Find the external power delivered in moving the rail with constant velocity v at time t (A = area of cross section of the bar, ρ = resistivity of the bar)



Ans: $\frac{2B^2 A v^3 \tan \theta}{\rho} t$

Sol: External power delivered = force on the rod due to magnetic field $\times$ velocity of the rod.

126. An LR circuit having a time constant of 50 ms is connected with an ideal battery of emf ε . Find the time elapsed before
 (a) the current reaches half its maximum values,
 (b) the magnetic field energy stored in the circuit reaches half its maximum value.

Ans: (a) $\frac{\ln 2}{20} s$ (b) $(-50 \times 10^{-3}) \ln(0.3) \text{ sec}$

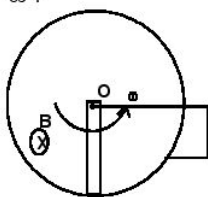
Sol: Use L-R circuit charging equation

127. A long solenoid that has 800 turns per meter carries a current $i = 3 \sin(400t) \text{ A}$. Find the electric field inside the solenoid at a distance 2 mm from the solenoid axis. Consider only the field tangential to a circle having its center on the axis of the solenoid.

Ans: $1.2 \times 10^{-3} \cos 400t \text{ v/m}$

Sol: Electric field $= \frac{r}{2} \frac{dB}{dt}$ where $r = 2\text{mm}$, $\frac{dB}{dt}$ = rate of change of magnetic field.

128. A metallic rod of length ℓ & resistance R is free to rotate about one of its ends over a smooth, rigid circular metallic frame of radius ℓ in an inward magnetic field of induction B . What torque should be applied by an external agent to rotate the rod with constant angular velocity ω ?

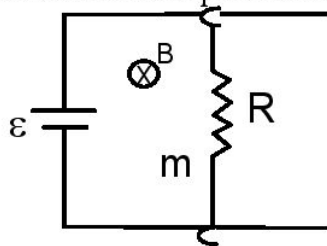


Ans: $\frac{B^2 \ell^4 \omega}{4R}$

Sol: Considering motion of the rod in horizontal plane only ; first find out current through the

rod, then force on a small element on the rod due to external magnetic field.

129. A sliding conducting bar of mass m , resistance R is released from rest. It starts sliding due to the current drawn from a battery of emf ε , in a steady inward magnetic field. Find the variation of its speed with time. Also find the terminal speed of the bar.



Ans: $\frac{E}{B\ell} \left(1 - e^{-\frac{B^2 \ell^2 t}{mR}} \right); \frac{E}{B\ell}$

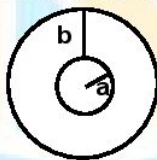
Sol: Write equation of motion of the rod at any instant.

130. An inductor of inductance 20 mH is connected across a charged capacitor of capacitance $5 \mu\text{F}$ & resulting L-C circuit is set oscillating at its natural frequency. The maximum charge q is $200 \mu\text{C}$ on the capacitor. Find the potential difference across the inductor, when the charge is $100 \mu\text{C}$.

Ans: 20 V

Sol: Use an L-C oscillating circuit relation.

131. The current in the inner coil is $I = 2t^2$. Find the heat developed in the outer coil between $t = 0$ and t seconds. The resistance of the inner coil is R and take $b \gg a$.

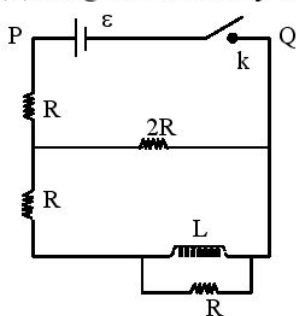


Ans: $\frac{4\mu_0^2 \pi^2 a^4 t^3}{b^2 R \cdot 3}$

Sol: Find flux enclosed by the smaller loop due to magnetic field of the larger loop.

132. In the figure shown is a R-L circuit connected with a cell of emf ε through a key k . If key k is closed find the current drawn by the battery

- (a) just after the key k is closed
(b) long after the key k is closed



Ans: (a) $\frac{\varepsilon}{2R}$ (b) $\frac{3\varepsilon}{5R}$

Sol: Remember an inductor is an open circuit at $t = 0$ and shorting at $t = \infty$

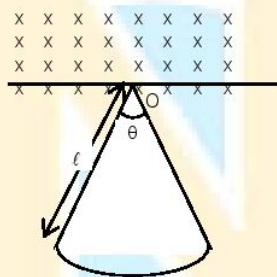
133. A very small circular loop of area $5 \times 10^{-4} \text{ m}^2$, resistance 2 ohm and negligible self inductance initially coplaner and concentric with a much larger fixed circular loop of radius 0.1 m. A constant current of 1.0 A is passed through the bigger loop. The smaller loop is rotated with constant angular velocity ω rad/sec about it's diameter. Calculate the (a) induced emf and (b) the induced current through the smaller loop as a function of time.

Ans: (a) Induced emf = $AB \sin \omega t = 3.14 \times 10^{-9} \sin \omega t$

(b) Induced current = $1.57 \times 10^{-9} \omega \sin \omega t$

Sol: Find flux enclosed by the smaller loop due to magnetic field of the larger loop.

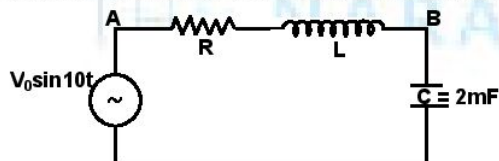
134. A wire in the form of a sector of radius ℓ and of angle ($\theta = \pi/4$) having a resistance R is free to rotate about an axis passing through point O and perpendicular to horizontal plane. A vertical magnetic field $B = -B_0 \hat{k}$ exists in the space. If the sector rotates with constant angular velocity so that Q Joules of heat is produced per revolution, find the constant angular velocity.



Ans: $\frac{8QR}{\pi B^2 \ell^4}$

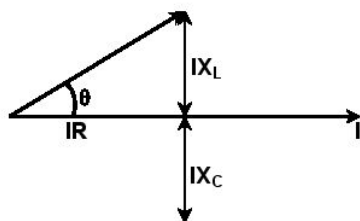
Sol: Induced current will flow while the loop enters in to field and again while it comes out. There will be no current when the loop is completely in the magnetic field.

135. In the given circuit, power factor between A and B is 0.5 and over all power factor of the circuit is 1. Find the value of R and L.



Ans: $R = \frac{\omega L}{0.5} = \frac{10 \times 5}{0.5} = 100 \Omega$

Sol: Phasor diagram of the given circuit will be



In first case

Power factor between A and B is 0.5

$$\cos\theta = \frac{\omega L}{R} = 0.5 \quad \dots(1)$$

In second case

Overall power factor is one so

$$X_L = X_C \quad \dots(2)$$

$$\omega L = \frac{1}{\omega C} \quad \text{or} \quad L = \frac{1}{100 \times 0.002} = 5H$$

put the value of L in (i)

$$R = \frac{\omega L}{0.5} = \frac{10 \times 5}{0.5} = 100\Omega$$

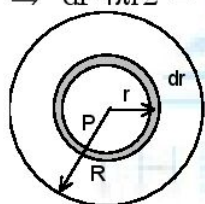
136. A uniform sphere has a mass M and radius R . Find the gravitational pressure P inside the sphere, as a function of the distance r from its centre.

Ans:
$$\frac{3}{8} G \frac{\{1 - r^2/R^2\} M^2}{\pi R^4}$$

Sol: Consider a layer of thickness dr at a distance r from the centre of the sphere.

Now force due to the layer $dF = 4\pi r^2 dP$

$$\Rightarrow dP 4\pi r^2 = \frac{G \left(\frac{4}{3} \pi r^3 \rho \right) (4\pi r^2 dr \rho)}{r^2}$$



(Where ρ is the mean density of sphere)

$$\text{or, } dP = G \frac{4}{3} \pi \rho^2 r dr$$

$$\therefore P = G \int_r^R \frac{4}{3} \pi \rho^2 r dr$$

$$\text{or, } P = G \frac{2\pi}{3} \rho^2 (R^2 - r^2)$$

$$= \frac{3}{8} G \frac{\{1 - r^2/R^2\} M^2}{\pi R^4} \quad [\because \rho = M / (4/3) \pi R^3]$$

137. What is the gravitational potential energy of a particle of mass m kept at a distance x from the centre of a disc of mass M on its axis? The radius of the disc is R .

Ans : $U = -\frac{2GMm}{R^2}(\sqrt{R^2 + x^2} - x)$

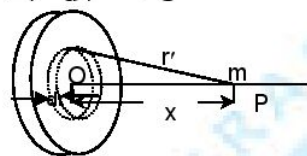
Solution: The gravitational potential of the differential ring is given as

$$dV = -G \frac{dM}{r'}$$

where $dM = \frac{M}{\pi R^2} \times 2\pi r dr = \frac{2Mr dr}{R^2}$

& $r' = \sqrt{r^2 + x^2}$

$$\Rightarrow dV = -G \left(\frac{2Mr dr}{R^2} \right) / \sqrt{r^2 + x^2}$$



$\Rightarrow$ The potential due to the entire disc at the point P is given as

$$V = \int dV = -\frac{2GM}{R^2} \int_0^R \frac{r dr}{\sqrt{r^2 + x^2}}$$

$$\Rightarrow V = -\frac{GM}{R^2} \int_0^R \frac{2r dr}{\sqrt{r^2 + x^2}}$$

Let $r^2 + x^2 = t^2$

$$\Rightarrow 2r dr = 2t dt$$

$$\therefore V = -\frac{GM}{R^2} \int \frac{2t dt}{t} = -\frac{2GM}{R^2} t$$

$$\Rightarrow V = -\frac{2GM}{R^2} \left[\sqrt{r^2 + x^2} \right]_0^R$$

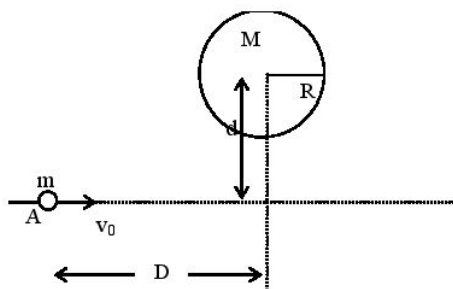
$$\Rightarrow V = -\frac{2GM}{R^2} \left[\sqrt{R^2 + x^2} - x \right]$$

The gravitational potential energy of the system

$$= U = mV$$

$$\Rightarrow U = -\frac{2GMm}{R^2} (\sqrt{R^2 + x^2} - x)$$

138. With what speed v_0 should a body be projected as shown in the figure, with respect to a planet of mass M so that it would just be able to graze the planet and escape? The radius of the planet is R .



$$v_0 = \sqrt{\frac{2GM \left(\frac{1}{R} - \frac{1}{\sqrt{D^2 + d^2}} \right)}{\left(\frac{d^2}{R^2} - 1 \right)}}$$

Ans: $v_0 =$

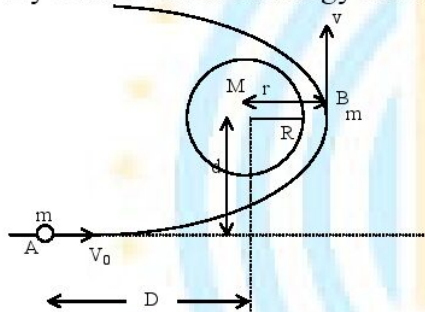
Sol: By conservation of angular momentum we obtain

$$mv_0 d = mvr$$

$$\Rightarrow v = \frac{mv_0 d}{mr}$$

$$= v_0 (d/r)$$

By conservation of energy at A and B, we have



$$\frac{1}{2} m v_0^2 - \frac{GMm}{\sqrt{D^2 + d^2}} = \frac{1}{2} m v^2 - \frac{GMm}{r}$$

When $r \approx R$ we obtain,

$$\frac{1}{2} m v_0^2 - \frac{GMm}{\sqrt{D^2 + d^2}} = \frac{1}{2} m \frac{v_0^2 d^2}{R^2} - \frac{GMm}{R}$$

$$\Rightarrow \frac{1}{2} m v_0^2 \left(\frac{d^2}{R^2} - 1 \right) = GMm \left[\frac{1}{R} - \frac{1}{\sqrt{D^2 + d^2}} \right]$$

$$\Rightarrow v_0 = \sqrt{\frac{2GM \left(\frac{1}{R} - \frac{1}{\sqrt{D^2 + d^2}} \right)}{\left(\frac{d^2}{R^2} - 1 \right)}}$$

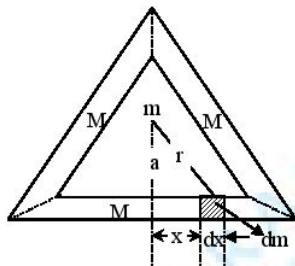
139. Three uniform rods, each of mass M and length ℓ , are connected to form an equilateral triangle in a gravity free space. Another small body of mass m is kept at the centroid. Find the minimum velocity V to be given to mass m so that it escapes the gravitational pull of the triangle.

Ans: $v = \left[\frac{12GM}{\ell} \ln \left(\frac{3+\sqrt{2}}{3\sqrt{2}} \right) \right]^{1/2}$

Sol: The gravitational potential energy between the point mass m and the elementary segment is given as

$$dU = -\frac{Gm(dm)}{r} = -Gm \left\{ \left(\frac{M}{\ell} \right) dx \right\} / \sqrt{a^2 + x^2}$$

$$= -\frac{GMm}{\ell} \frac{dx}{\sqrt{a^2 + x^2}}$$



$\therefore$ The total gravitational potential energy associated with the system

$$= U = 3 \int dU$$

$$= -\left(\frac{3GMm}{\ell} \right) 2 \int_0^{\ell/2} \frac{dx}{\sqrt{a^2 + x^2}}$$

$$= -\frac{6GMm}{\ell} \left[\ln \left| \frac{x}{a} + \frac{\sqrt{a^2 + x^2}}{a} \right| \right]_0^{\ell/2}$$

$$\Rightarrow U = -\frac{6GMm}{\ell} \ln \left| \frac{\ell/2 + \sqrt{a^2 + (\ell/2)^2}}{a} \right|$$

$$U = -\frac{6GMm}{\ell} \ln \left| \frac{\ell + \sqrt{\ell^2 + 4a^2}}{2a} \right|$$

For minimum velocity $|U| = \frac{1}{2}mv^2$

$$\therefore v = \left[\frac{12GM}{\ell} \ln \left| \frac{\ell + \sqrt{\ell^2 + 4a^2}}{2a} \right| \right]^{1/2}$$

Putting $a = \frac{\ell}{2\sqrt{3}}$,

$$v = \left[\frac{12GM}{\ell} \ln \left(\frac{3+\sqrt{2}}{3\sqrt{2}} \right) \right]^{1/2}$$

140. A planet of mass m moves along an ellipse around the sun so that its maximum and minimum distance from the sun is equal to r_1 and r_2 respectively. Find the angular momentum of this plane relative to the centre of the sun.

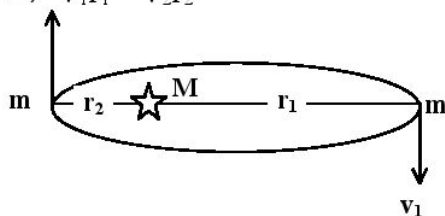
Ans: $L = mvr_1 = m \sqrt{\frac{2GM r_1 r_2}{r_1 + r_2}}$

Sol: The angular momentum $= L = mvr_1$... (1)

Conservation of angular momenta at 1 and 2;

$$mv_1 r_1 = mv_2 r_2$$

$$\Rightarrow v_1 r_1 = v_2 r_2 \quad \dots (2)$$



Conservation of energy at 1 and 2;

$$\begin{aligned} \frac{1}{2} mv_1^2 - \frac{GMm}{r_1} \\ = \frac{1}{2} mv_2^2 - \frac{GMm}{r_2} \quad \dots (3) \end{aligned}$$

Using (2) and (3) we obtain

$$v_1 = \sqrt{\frac{2GM r_2}{r_1(r_1 + r_2)}}$$

$$\therefore L = mvr_1 = m \sqrt{\frac{2GM r_1 r_2}{r_1 + r_2}}$$

141. An artificial satellite (mass m) of a planet (mass M) revolves in a circular orbit whose radius is n times the radius R of the planet. In the process of motion, the satellite experiences a slight resistance due to cosmic dust. Assuming the force of resistance on satellite to depend on velocity as $F = av^2$ where ' a ' is a constant, calculate how long the satellite will stay in the orbit before it falls onto the planet's surface.

Ans : $t = \frac{m\sqrt{R}(\sqrt{n}-1)}{a\sqrt{GM}} = (\sqrt{n}-1) \frac{m}{a\sqrt{gR}}$.

Sol: Air resistance $F = -av^2$ where orbital velocity $v = \sqrt{\frac{GM}{r}}$; r = the distance of the satellite from planet's center.

$$\Rightarrow F = -\frac{GMa}{r}$$

The work done by the resistance force

$$= dW = F \cdot dx = [F v dt]$$

$$= \left[\frac{GMa}{r} \sqrt{\frac{GM}{r}} dt \right] = \left[\frac{(GM)^{3/2} a}{r^{3/2}} dt \right] \quad \dots (1)$$

$\Rightarrow$ The loss of energy of the satellite $= dE$

$$\therefore \frac{dE}{dr} = \frac{d}{dr} \left[-\frac{GMm}{2r} \right] = \frac{GMm}{2r^2}$$

$$\Rightarrow dE = \frac{GMm}{2r^2} dr \quad \dots (2)$$

Since $dE = -dW$ (work energy theorem)

$$-\frac{GMm}{2r^2} dr = \frac{(GM)^{3/2} a}{r^{3/2}} dt$$

$$t = -\frac{m}{2a\sqrt{GM}} \int_{nR}^R \frac{dr}{\sqrt{r}}$$

$$\Rightarrow t = \frac{m\sqrt{R}(\sqrt{n}-1)}{a\sqrt{GM}} = (\sqrt{n}-1) \frac{m}{a\sqrt{gR}}$$

142. A satellite of mass 2×10^3 kg has to be shifted from an orbit of radius $2R$ to another radius of $3R$, where R is the radius of earth. Calculate the minimum energy required. $R = 6400$ km and $g = 10 \text{ m/s}^2$.

Ans: $\Delta E = \frac{mgR}{12} = 1.067 \times 10^{10} \text{ J}$

Sol: Total mechanical energy of a satellite in a circular orbit of radius r

$$= -\frac{1}{2} \frac{GMm}{r}$$

$$E_1 = \text{energy in first orbit} = -\frac{1}{2} \frac{GMm}{2R}$$

$$E_2 = \text{energy in second orbit} = -\frac{1}{2} \frac{GMm}{R}$$

$$\Delta E, \text{ energy (minimum) required} = E_2 - E_1 = \frac{1}{2} \frac{GMm}{R} \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$\Rightarrow \Delta E = \frac{GMm}{12R} = \frac{mgR^2}{12R} \quad (GM = gR^2)$$

$$\Rightarrow \Delta E = \frac{mgR}{12} = 1.067 \times 10^{10} \text{ J}$$

143. Two satellites A and B, of equal mass, move in the equatorial plane of earth close to the earth's surface. Satellite A moves in the same direction as that of the rotation of the earth while satellite B moves in the opposite direction. Determine the ratio of the kinetic energy of B to that of A in the reference frame fixed to earth.

Ans: $\frac{KE_1}{KE_2} = 1.265$

Sol: The orbital speed of a satellite very close to earth $= V_0 = \sqrt{\frac{GM}{R}} = \sqrt{g_0 R}$

$$\text{The peripheral speed of earth} = V_e = R\omega_e = \frac{2\pi}{T_e} R$$

$\Rightarrow$ The velocities of the satellites relative to earth are given by

$$V_{r_1} = \frac{2\pi}{T_e} R + \sqrt{g_0 R} \quad \text{and} \quad V_{r_2} = \sqrt{g_0 R} - \frac{2\pi}{T_e} R$$

Positive and Negative sign are for the satellites orbiting from east to west and west to east respectively because earth rotates from west to east

$$\Rightarrow \frac{KE_1}{KE_2} = \frac{\frac{1}{2} m V_{r_1}^2}{\frac{1}{2} m V_{r_2}^2} = \left(\frac{V_{r_1}}{V_{r_2}} \right)^2 = \frac{\left(\frac{2\pi}{T_e} R + \sqrt{g_0 R} \right)^2}{\left(-\frac{2\pi}{T_e} R + \sqrt{g_0 R} \right)^2}$$

Putting $R = 6.4 \times 10^6 \text{ m}$, $g_0 = 9.8 \text{ m/sec}^2$ and $T_e = 86400 \text{ sec}$,

$$\frac{KE_1}{KE_2} = 1.265$$



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